



**Enhancing lithium ion conductivity in the solid electrolyte
 $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ for all solid-state batteries**

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Author: [REDACTED]

Student ID: [REDACTED]

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Supervisor: Ming Li

Summary

Lithium-ion batteries (LIBs) are widely used in electric vehicles (EVs) due to their high energy density and long lifetime. However, EVs suffer from safety issues because of the flammable liquid electrolytes used in LIBs. Solid-state electrolytes (SSEs) can effectively solve this issue with enhanced stability and safety, but their ionic conductivity is lower than that of liquid electrolytes presently. Among numerous varieties of solid electrolytes, garnet-type $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ (LLZO) has received extensive attention due to its relatively high ionic conductivity and its potential for future development. However, its ionic conductivity is still inferior to that of liquid electrolyte batteries. Thus, this project focuses on enhancing the lithium ionic conductivity of LLZO. Elemental doping and advanced sintering techniques will be implemented, which are two typical methods for LLZO ionic conductivity improvement. It is worth investigating because it may substantially improve the performance of SSEs and the safety of EVs.

This project aims to enhance the lithium ionic conductivity in the solid electrolyte LLZO by doping with Bi^{3+} and Sb^{5+} elements. Additionally, spark plasma sintering (SPS) is employed to further enhance the ionic conductivity. The objectives include generating phase-pure LLZO samples and obtaining dense ceramics with 95% or even higher relative density and up to $10^{-4} \text{ S cm}^{-1}$ ionic conductivity at room temperature. Fifteen $\text{Li}_{6.6}\text{La}_{3-y}\text{Bi}_y\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ solid electrolyte samples were synthesised with various Bi doping ratios, sintering temperatures, and sintering techniques. Four groups of experiments were conducted to investigate the effect of Bi doping amount under the conventional sintering method or under SPS, the effect of the sintering temperature, and the effect of the advanced sintering technique SPS on the performance and characteristics of LLZO solid electrolytes.

Among fifteen LLZO samples, $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ sintered at 1030°C for 10 h achieved the highest conductivity of $1.75 \times 10^{-4} \text{ S cm}^{-1}$. This suggests that the optimal doped Bi content for LLZO sintered by the conventional method may be 0.3. All of the relative densities of the SPS sintered samples were significantly higher than those of the conventionally sintered samples. The highest relative density was 94.9% which was possessed by $\text{Li}_{6.6}\text{La}_{2.85}\text{Bi}_{0.15}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ sintered at SPS 950°C for 5 minutes. Also, this sample obtained the second highest ionic conductivity of $8.27 \times 10^{-5} \text{ S cm}^{-1}$, demonstrating that 0.15 is the optimal Bi doping concentration for SPS sintered LLZO. Furthermore, the ionic conductivity increased as the conventional sintering temperature increased from 1000°C to 1030°C , indicating that 1030°C is the optimal conventional sintering temperature.

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NOMENCLATURE

Symbol	Quantity	Unit
ρ	density	g/cm^3
m	mass	g
V	volume	cm^3
t	thickness	cm
D	diameter	cm
m_1	dry weight	g
m_2	wet weight	g
ρ_{water}	density of water	g/cm^3
ρ_{th}	theoretical density	g/cm^3
N	Number of atoms per unit cell	1
A_w	Atomic weight	amu
V_c	volume of unit cell	nm^3
N_A	Avogadro's number	1/mol
ρ_{re}	relative density	%
n	integer	1
λ	wavelength of x-rays	m
d	crystal plan spacing	Å
θ	diffraction angle	°
σ	ionic conductivity	S cm^{-1}
R	resistance	Ω
S	area	cm^2
A	pre-exponential factor	1
E_a	activation energy	eV
K_B	Boltzmann constant	J/K
T	Temperature	K

ACRONYMS

Acronym	Meaning
EV	Electric vehicles
LIB	Lithium-ion battery
SSE	Solid-state electrolyte
LLZO	$\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$
SPS	Spark plasma sintering
FAST	Field-assisted sintering technology
HP	Hot pressing
t-LLZO	Tetragonal phase $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$
c-LLZO	Cubic phase $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$
PEO	Polyethylene oxide
DC	Direct current
LLZBO	$\text{Li}_{6.8}\text{La}_3\text{Zr}_{1.8}\text{Bi}_{0.2}\text{O}_{12}$
XRD	X-ray diffraction
EIS	Electrochemical impedance spectroscopy
SEM	Scanning electron microscopy
AC	Alternating current

1. INTRODUCTION.

1.1. Background and Context

The global energy crisis is becoming increasingly problematic, and new technologies are required to replace conventional fossil fuels with cleaner and more efficient energy sources [1]. Electric vehicles (EVs) are gaining popularity due to their eco-friendly characteristics, long lifespan, and low usage cost. They can mitigate the environmental impact of fossil fuels significantly [2]. Thus, lithium-ion batteries (LIBs) have significant potential in energy storage, and LIBs are the primary energy sources for EVs due to numerous advantages, such as their high energy density, high voltage, lightweight and small volume, low self-discharging, non-memory effect, and extended lifecycle [3-5]. They have higher energy densities than lead-acid, alkaline, and Ni-Cd batteries [6]. Currently, conventional liquid electrolytes with high ionic conductivity (up to $10^{-2} \text{ S cm}^{-1}$) are used in commercial LIBs for EVs. It can result in rapid ion movements and beneficial battery performance [7]. Nonetheless, EVs have safety issues due to the flammable liquid electrolytes present in LIBs, which cause dendritic growth and thermal runaway issues [8, 9]. Additionally, there is also the risk of liquid electrolytes leaking. Figure 1 shows the car fire caused by LIBs.



Figure 1. Tesla car fire started in the battery [10].

Within this context, solid-state electrolytes (SSEs) have been considered promising replacements for liquid electrolytes. Due to the lack of flammable liquid electrolytes commonly used in LIBs, the primary advantage of SSEs is enhanced safety. In addition, compared to liquid electrolytes, SSEs have slower solid-state reactivity, which could result in an extended lifetime and cyclability. It is proven by solid-state microbatteries that can operate for more than 10,000 cycles. Also, no significant effects of ageing are observed in solid electrolytes, as this phenomenon is associated with liquid electrolytes [11]. Figure 2 depicts a comparison of liquid electrolyte batteries and solid electrolyte batteries. As the critical component of batteries, SSEs are expected to have high ion conductivity. Nevertheless, the challenge of SSEs currently is that their ionic conductivity is still orders of magnitude below that of liquid electrolytes in general [9].

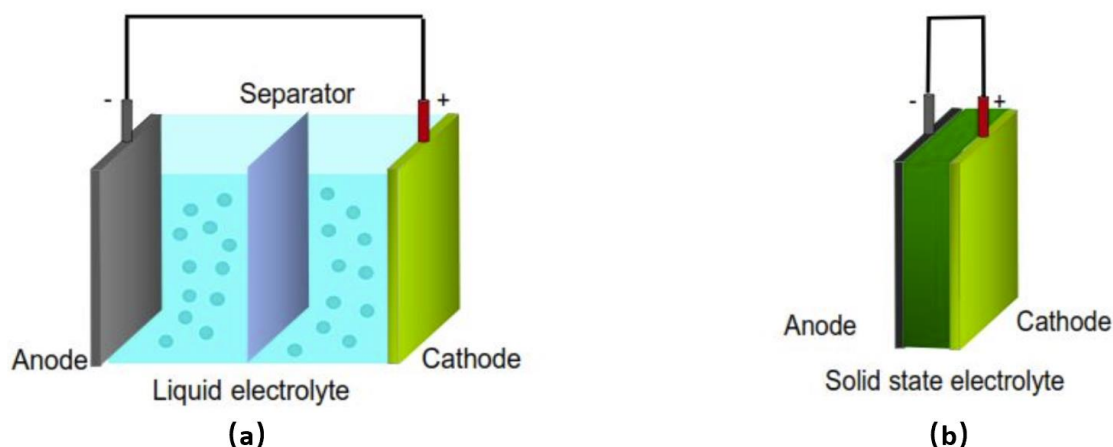


Figure 2. Schematic illustration of (a) LIBs containing the liquid electrolyte; (b) all solid-state battery [12].

SSEs can be broadly categorised into two groups: inorganic solid electrolytes, represented by oxides and sulphides, and organic solid electrolytes, represented by polymers such as polyethylene oxide (PEO) [13]. Compared to polymer electrolytes, inorganic solid electrolytes have a lower shear modulus, larger ionic conductivity, and a wider range of operating temperature [11, 14]. Thus, the research on inorganic solid electrolytes is more intensive. Among inorganic solid electrolytes, garnet-type $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ (LLZO) is one of the most promising SSE due to its relatively high conductivity, which reaches 10^{-3} to $10^{-4} \text{ S cm}^{-1}$ [15]. In addition, it also has favourable chemical stability and thermal performance [7, 9, 16]. However, the ionic conductivity of LLZO is still considerably lower than that of liquid electrolytes. The effects of crystal structure, the sintering temperature and the grain boundary account for this result [7]. Many approaches can be applied to improve the ionic conductivity of LLZO.

Typically, two methods can be used to increase the ionic conductivity of LLZO. They are elemental doping and advanced sintering techniques. Firstly, it can be optimised by doping elements. LLZO has two kinds of polymorphs, including the tetragonal phase LLZO (t-LLZO) and the cubic phase LLZO (c-LLZO). The ionic conductivity of c-LLZO is larger than that of t-LLZO, but it is not stable at room temperature. Elemental doping can effectively stabilise c-LLZO and result in enhanced ionic conductivity. According to previous research, multiple doping elements can be used, such as Bi^{3+} , Sb^{5+} , Al^{3+} , Ga^{3+} and Ti^{4+} [7]. The general formula is $\text{Li}_{7-x}\text{La}_3\text{Zr}_{2-x}\text{M}_x\text{O}_{12}$ (M is the doping element). In addition to single doping, co-doping can further improve the ionic conductivity [16]. Furthermore, the sintering temperature and the moisture resistance play important roles in the performance of LLZO. Thus, the properties of SSEs can also be improved by optimising the manufacturing process. Compared to the conventional sintering method, some advanced sintering techniques can be used in the sintering process, such as spark plasma sintering (SPS), field-assisted sintering technology (FAST), and hot pressing (HP) [17, 18]. Currently, SPS is considered a promising method. It is a low voltage DC pulse current activated, pressure assisted sintering technique that can produce materials with high densification [19]. Moreover, it can not only reduce the sintering

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2. LITERATURE REVIEW

2.1. Working principle of Lithium-ion battery

LIB is a type of commonly used rechargeable batteries. The LIB consists of four parts: electrolyte, cathode, anode, and separator. The function of the cathode and anode is assisting the storage and release of the energy of the LIB. When the LIB is charged, Li^+ forms on the cathode and diffuses through the electrolytes to the anode. Simultaneously, electrons are transferred through the external circuit in the identical orientation. When the LIB is discharged, Li^+ moves from the anode to the cathode. The separator is used to block the electrodes to avoid internal short-circuits while allowing the flow of Li^+ . The function of the electrolyte is to carry Li^+ ions instead of conducting electrons. The schematic of the LIB working principle is demonstrated in Figure 3. Taking $\text{LiCoO}_2/\text{graphite}$ LIB as an example, the reactions of the charging and discharging processes are [21]:

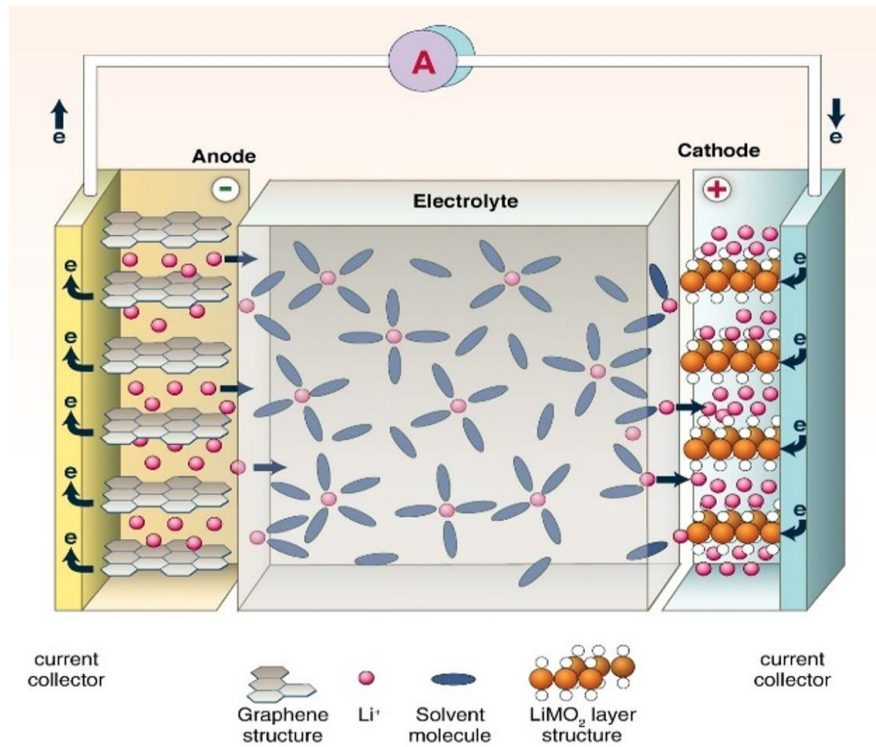
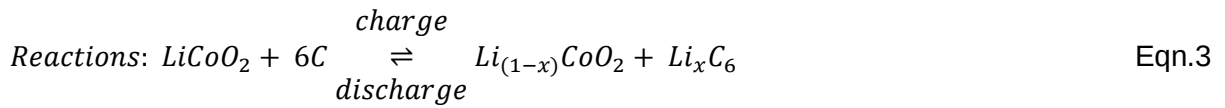
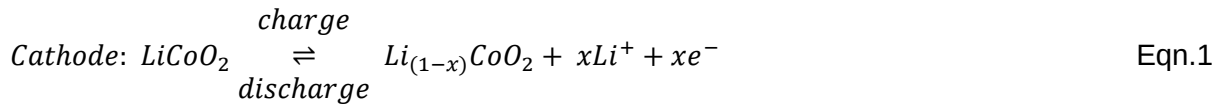


Figure 3. Schematic of the working principle of a LIB [22].

2.2. Crystal Structure of garnet-type LLZO

LLZO solid electrolytes have two different crystal structures based on the location of Li-ion in the lattice. They are tetragonal phase (t-LLZO, space group $I4_1/acd$) (Figure 4a) and cubic phase (c-LLZO, space group $Ia\bar{3}d$) (Figure 4b) [23]. There is no structural distinction between these two phases. The difference is in the distribution of Li atoms, which is the main determination of the ionic conductivity of LLZO. In t-LLZO, Li^+ occupies tetrahedral 8a, octahedral 16f and 32g these three different lithium sites (shown in Figure 4a). Li(1) belongs to 8a tetrahedral site lined with four O atoms, and Li(2) and Li(3) is in 16f and 32g octahedral sites which are bonded with six O atoms. In c-LLZO, Li^+ only occupies 24d tetrahedral and 96h octahedral lithium sites. Thus, Li^+ has more available sites for transportation in c-LLZO than in t-LLZO in terms of site occupancy [24]. Consequently, the ionic conductivity of c-LLZO is approximately three orders of magnitude higher than that of t-LLZO [25]. However, pure c-LLZO tends to shift to the tetragonal structure due to its instability [26, 27]. Additionally, the reason for the decreased stability is that pure c-LLZO tends to absorb H_2O and CO_2 in the air, which makes it generate the heterogeneous phase and the structure degradation due to the H^+/Li^+ exchange [28]. As an effective solution to this problem, LLZO can be doped with elements that have a similar ionic radius to improve the stability of the cubic phase lattice and the ionic conductivity. This is because doping elements can decrease the content of Li and augment the concentration of Li vacancies [23].

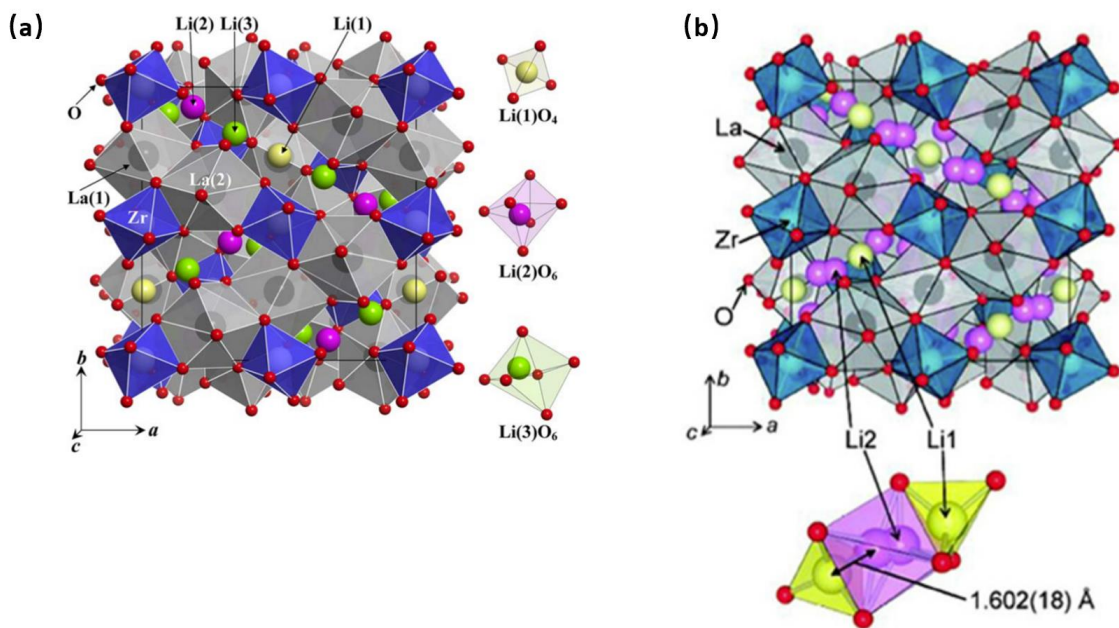


Figure 4. Crystal Structure of LLZO: (a) t-LLZO [29]; (b) c-LLZO [30].

2.3. Enhance ionic conductivity by elemental doping

It has been demonstrated that various elements can be doped into LLZO to improve its ionic conductivity. Elemental doping can stabilise the c-LLZO and improve ionic conductivity. The cell size of c-LLZO can be enlarged or shrunk due to the doping element, meaning that the

doping element increases the migration of Li^+ from one position to another. Additionally, The Li content can be adjusted by the doping element, resulting in a more disordered lattice arrangement that can enhance the stability of the cubic structure of LLZO. These are the reasons why ionic conductivity can be increased by elemental doping [24].

Depending on the distinct doping site, elemental doping can be divided into three types, including Li-site substitution, La-site substitution, and Zr-site substitution. Zr-site substitution is investigated the most because Zr-doped elements enter the lattice without side reactions, and it has no effect on Li^+ distribution and has no interference with Li^+ transportation. In contrast, the present research illustrates that La-site substitution is less effective than that of Zr-site [23]. The ideal doping element should simultaneously reduce the activation barrier for the formation of the compound and enhance the densification during the sintering process [9]. In this project, Bi^{3+} and Sb^{5+} are selected as doping elements. Zr^{4+} (larger ionic radius of 0.72 Å in VI coordination) is substituted by Sb^{5+} (smaller ionic radius of 0.61 Å in VI coordination) [31]. Bi^{3+} can be used as a substitute on the La^{3+} site. Bi^{3+} and Sb^{5+} are selected because they have lone pair electrons and are prone to form weak Bi-O bonds and Sb-O bonds, which contribute to the migration of Li^+ .

It is reported that the ionic conductivity of $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ was $5 \times 10^{-4} \text{ S cm}^{-1}$ at 25 °C [32]. By contrast, for the doping element Sb^{5+} , the ionic conductivity of $\text{Li}_{7-x}\text{La}_3\text{Zr}_{2-x}\text{Sb}_x\text{O}_{12}$ ($x=0.2-1.0$) made by the solid-state reaction method (sintered at 1100 °C for 24 h) can be optimised by doping with Sb^{5+} , and the maximum ionic conductivity was obtained at room temperature at a concentration of approximately $x=0.4$. $\text{Li}_{6.6}\text{La}_3\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ had an ionic conductivity of $7.7 \times 10^{-4} \text{ S cm}^{-1}$ at 30 °C and the relative density reached 91%, which is better than that of undoped LLZO (less than 78%) [31]. For the doping element Bi^{3+} , research shows that the ionic conductivity of $\text{Li}_{6.25}\text{La}_3\text{Zr}_{1.25}\text{Bi}_{0.75}\text{O}_{12}$ ($\text{Li}_{7-x}\text{La}_3\text{Zr}_{2-x}\text{Bi}_x\text{O}_{12}$, $x = 0.75$) (sintered at 900 °C for 10 h) at 27 °C is more than 100 times larger than that of the undoped sample (from $10^{-6} \text{ S cm}^{-1}$ to $2.0 \times 10^{-4} \text{ S cm}^{-1}$) [9].

In addition, both single doping and co-doping will be attempted in this project. Co-doping may result in higher ionic conductivity due to structural changes than single doping. The ionic conductivity of Sb-Ba co-doping is $1.53 \times 10^{-4} \text{ S cm}^{-1}$ $\text{Li}_{6.945}\text{La}_{2.98}\text{Ba}_{0.02}\text{Zr}_{1.925}\text{Sb}_{0.075}\text{O}_{12}$ which is higher than that of single Sb-doped LLZO ($\text{Li}_{6.925}\text{La}_3\text{Zr}_{1.925}\text{Sb}_{0.075}\text{O}_{12}$, $\sigma=1.31 \times 10^{-4} \text{ S cm}^{-1}$) [33]. It is also reported that adding Al into Bi-doped LLZO can significantly enhance the ionic conductivity. The ionic conductivity of $\text{Li}_{6.8}\text{La}_3\text{Zr}_{1.8}\text{Bi}_{0.2}\text{O}_{12}$ (LLZBO) without Al doping is only $2.6 \times 10^{-5} \text{ S cm}^{-1}$ (sintered at 1100 °C for 9 h), while doping with 1.25 mol.% Al_2O_3 in the original composition can increase the ionic conductivity to $6.3 \times 10^{-5} \text{ S cm}^{-1}$ [34]. The above data obtained from the previous research is summarised in Table 1.

Table 1. Previously reported LLZO data conclusion

Composition	Sintering condition	Relative density (%)	Ionic conductivity (S cm^{-1})	References
$\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$	-	-	5×10^{-4} at 25 °C	[32]
$\text{Li}_{6.6}\text{La}_3\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	1100 °C 24 h	91	7.7×10^{-4} at 30 °C	[31]
$\text{Li}_{6.25}\text{La}_3\text{Zr}_{1.25}\text{Bi}_{0.75}\text{O}_{12}$	900 °C 10 h	83	2.0×10^{-4} at 27 °C	[9]
$\text{Li}_{6.925}\text{La}_3\text{Zr}_{1.925}\text{Sb}_{0.075}\text{O}_{12}$	1180 °C 6 h	91	1.31×10^{-4} at 30 °C	[33]
$\text{Li}_{6.945}\text{La}_{2.98}\text{Ba}_{0.02}\text{Zr}_{1.925}\text{Sb}_{0.075}\text{O}_{12}$		95	1.53×10^{-4} at 30 °C	
$\text{Li}_{6.8}\text{La}_3\text{Zr}_{1.8}\text{Bi}_{0.2}\text{O}_{12}$	1100 °C 9 h	-	2.6×10^{-5} at room temperature	[34]
$\text{Li}_{6.8}\text{La}_3\text{Zr}_{1.8}\text{Bi}_{0.2}\text{O}_{12}$ -1.25% Al		-	6.3×10^{-5} at room temperature	

2.4. Synthesis of LLZO and sintering techniques

To obtain LLZO with enhanced performance, the synthesis method and the sintering condition are also crucial. In addition to the traditional solid-state reaction method, some advanced sintering techniques that could be applied to enhance pellet densification, such as hot pressing (HP), field-assisted sintering technology (FAST), and spark plasma sintering (SPS). Solid electrolytes with high relative density may have lower grain boundary resistance, resulting in lower activation energy and higher lithium ionic conductivity. Additionally, a high relative density could prevent the formation of dendrites, thereby enhancing safety and mechanical strength [24].

The traditional method of synthesising LLZO is the solid-state reaction method. The mixed metal oxides are sintered by this method at temperatures ranging from 800 °C to 1500 °C. This heat-treatment method is suggested to have two stages to avoid the formation of impurity compounds or the presence of unreacted precursor reactants. The first step is calcination at moderate temperatures, and the second is sintering at high temperatures. After the first stage, the pellets should be grounded again to discover the unreacted composition. In this way, single-phase LLZO with high purity can be fabricated [25].

HP is a sintering technique that simultaneously applies heat and uniaxial pressure. Compared to conventional sintering, the uniaxial pressure can facilitate pellet densification at a lower temperature and shorter sintering duration. This can lead to a lower Li loss, which can considerably improve the accuracy of the lithium ionic conductivity, as the lithium ionic conductivity is strongly dependent on the Li content [24, 35].

FAST is a kind of advanced sintering technique utilised at a high heating rate. This process resembles HP, with the addition of a direct current flowing through the tool in addition to the uniaxial pressure. There is an external electrical circuit that generates supplementary heat. This sintering technique can also reduce the sintering time, causing the reduction of Li^+ loss. A further advantage is that it integrates the synthesis of raw materials and sintering into a single process, preventing the time-consuming intermittent grinding and repeated sintering for a long duration [24].

Furthermore, SPS is similar to FAST. The difference is that SPS uses the pulsed direct current (DC). This technique can effectively speed up the process, reduce the sintering time, and densify the powders in one step [36]. It involves using a high electrical current to produce spark plasma between the powders of LLZO, followed by uniaxial compression to achieve high densification. It can also reduce Li loss, generating finely distributed particles of nanoscale dimensions and decreasing the grain boundary resistance [24]. It also significantly reduces the sintering temperature, which may reduce the formation of secondary phases. Additionally, high pressure (50 to 75 MPa) can be used during the sintering process to improve the physical and chemical stability of LLZO [25]. To better explain the reaction inside the SPS machine, the schematic diagram of the SPS (Figure 5a), the profile of the graphite die (Figure 5b), and the sample powder reaction in SPS compacted into pellets (Figure 5c) are depicted.

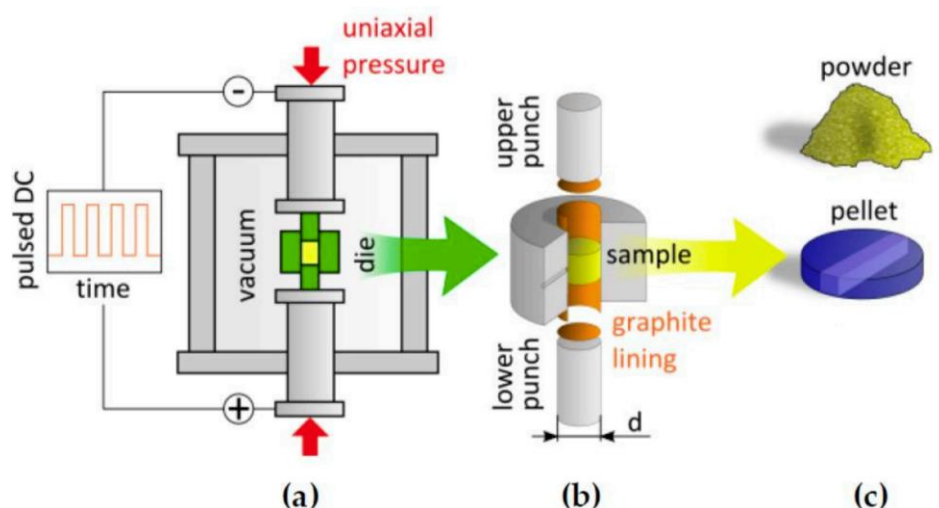


Figure 5. (a) Setup of SPS system; (b) cross-section view of SPS die; (c) the reaction inside SPS from powder to pellet. [37]

2.5. Research related to this project

In the above literature review, c-LLZO and t-LLZO crystal structures were explained. It demonstrates the microscopic causes for the greater lithium ionic conductivity of c-LLZO compared to t-LLZO. In t-LLZO, the positions and vacancies of Li^+ are organised, making it challenging for ions to transmit between adjacent positions. In consequence, the ionic conductivity of c-LLZO is approximately three orders of magnitude higher than that of t-LLZO [25]. However, pure c-LLZO tends to shift to the tetragonal structure due to its instability at high temperatures [26, 27]. Therefore, pure c-LLZO is anticipated to be prepared and stabilised in this project. Elemental doping and advanced sintering techniques were two effective methods to stabilise c-LLZO and increase the ionic conductivity. In this project, LLZO samples will be partial replaced by Bi^{3+} on La sites and Sb^{5+} on Zr sites to stabilise the cubic phase and increase the lithium ionic conductivity. In addition, SPS technique is employed to further enhance the ionic conductivity.

3. METHODOLOGY

3.1. The synthesis of $\text{Li}_{7-x}\text{La}_{3-y}\text{Bi}_y\text{Zr}_{2-x}\text{Sb}_x\text{O}_{12}$ sample

3.1.1. The synthesis of $\text{Li}_{7-x}\text{La}_{3-y}\text{Bi}_y\text{Zr}_{2-x}\text{Sb}_x\text{O}_{12}$ powder

Li_2CO_3 (Sigma-Aldrich, 99.99%), La_2O_3 (Sigma-Aldrich, 99.9%), ZrO_2 (Sigma-Aldrich, 99%), and Sb_2O_3 (Alfa Aesar, 99.6%), Bi_2O_3 (Acros.orgnics, 99.9%) were the raw materials for the fabrication of $\text{Li}_{7-x}\text{La}_{3-y}\text{Bi}_y\text{Zr}_{2-x}\text{Sb}_x\text{O}_{12}$. Solid-state reaction method was employed for the synthesis of LLZO samples. In the beginning, Li_2CO_3 , Sb_2O_3 and Bi_2O_3 powders were deposited in crucibles (Figure 6a) and dried at 300 °C for 3 h in the Carbolite Gero furnace CWF 1200 (Figure 6b), whereas La_2O_3 , ZrO_2 powders were dried at 1000 °C for 10 h. After cooling, the powders were weighed in the desired stoichiometry using the Mettler-Toledo analytical balance (Figure 6c). Particularly, 10 wt% excess Li_2CO_3 was required to compensate for the volatilization of Li during the sintering process. The powders were then manually mixed and ground with acetone as solvent using a mortar and pestle (Figure 6d) for approximately 30 minutes. After being milled, the powders were pressed into pellets in a die (Figure 6e) using the uniaxial press (Figure 6f) at 0.5 tons and held for about 1 minute. The pellets were then put in the crucible and calcined at 900 °C for 6 h in the furnace. Following calcination, the powders were cooled to room temperature. Then, a small amount of the calcined powder was taken and sent for powder X-ray diffraction (XRD).

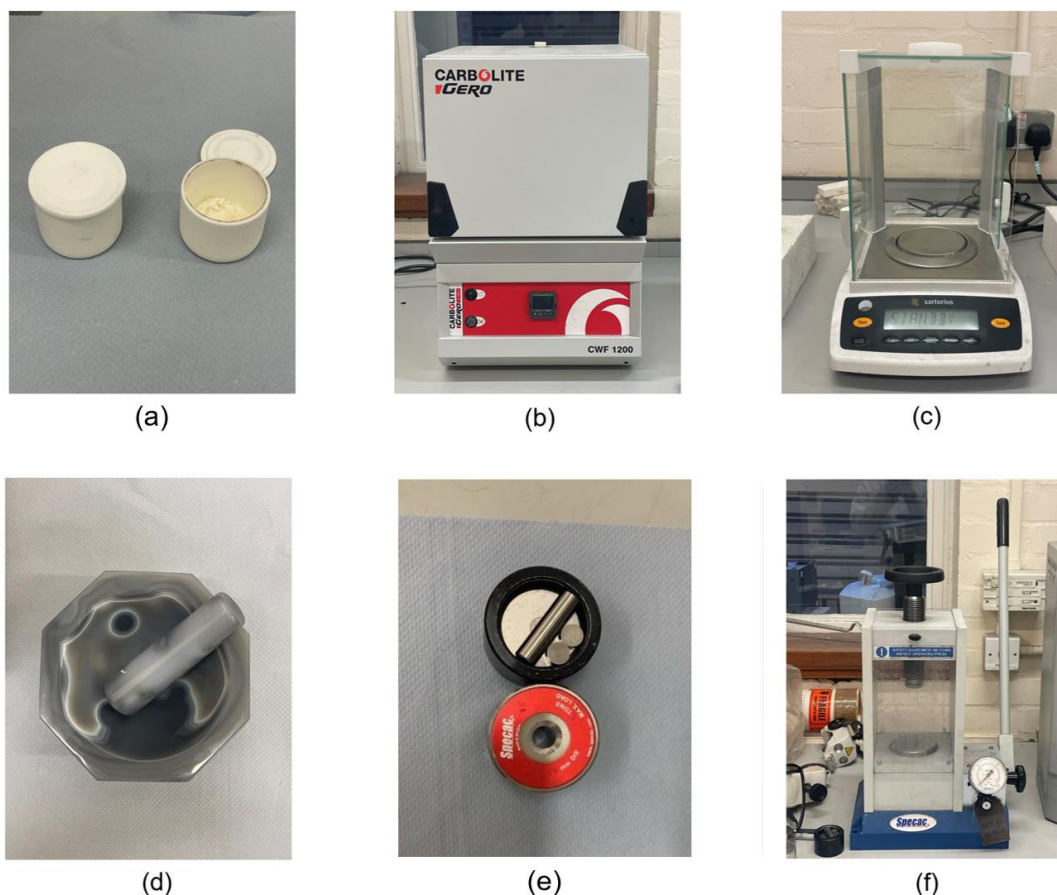


Figure 6. Images of the equipment and rigs: (a) crucibles; (b) furnace; (c) balance; (d) mortar and pestle; (e) die; (f) uniaxial press.

3.1.2. The synthesis of $\text{Li}_{7-x}\text{La}_{3-y}\text{Bi}_y\text{Zr}_{2-x}\text{Sb}_x\text{O}_{12}$ pellet

After finishing the synthesis of $\text{Li}_{7-x}\text{La}_{3-y}\text{Bi}_y\text{Zr}_{2-x}\text{Sb}_x\text{O}_{12}$ powder, the calcined pellets were secondary mixed and milled manually under dry conditions for approximately 40 minutes. The main purpose of the first grinding was to thoroughly mix the raw materials, while the second grinding was to make the powders as fine as possible. The resultant powders were then divided to sinter by different sintering techniques. Some of the resultant powders were sintered by solid-state reaction method. The powders were pressed into pellets via the 10 mm die under 3 tons for about 5 minutes of holding time. Then, the pellets were sintered in the furnace at an appropriate temperature (1000 °C or 1030 °C) for a suitable duration (10 h).

The remaining resultant powders were sintered by SPS. The graphite paper was first placed in the inner face of the graphite die (Figure 7a), and then the powders were poured between two graphite punches. Two circular graphite sheets were used to separate the powder and the punches. Afterwards, the graphite cloth was used to cover the graphite die. The prepared samples were then put into the chamber of the SPS equipment SPS-630SX (SUGA Co., Ltd.) (Figure 7b). Then, the SPS chamber was evacuated to a low vacuum status (at approximately 3 Pa). A radiation thermometer was pointed at the position of the powders to monitor the temperature in real-time. Various temperatures (including 750°C, 800°C, 950°C and 1000°C) were tried to find the optimum temperature. The holding time was 3 to 5 minutes. After pelletisation, the samples were extracted, and the carbon adhering to them was removed. Finally, the samples were placed in the furnace for annealing at 800°C for 8 hours. The flow chart of the whole process of the synthesis of LLZO is shown in Figure 8.

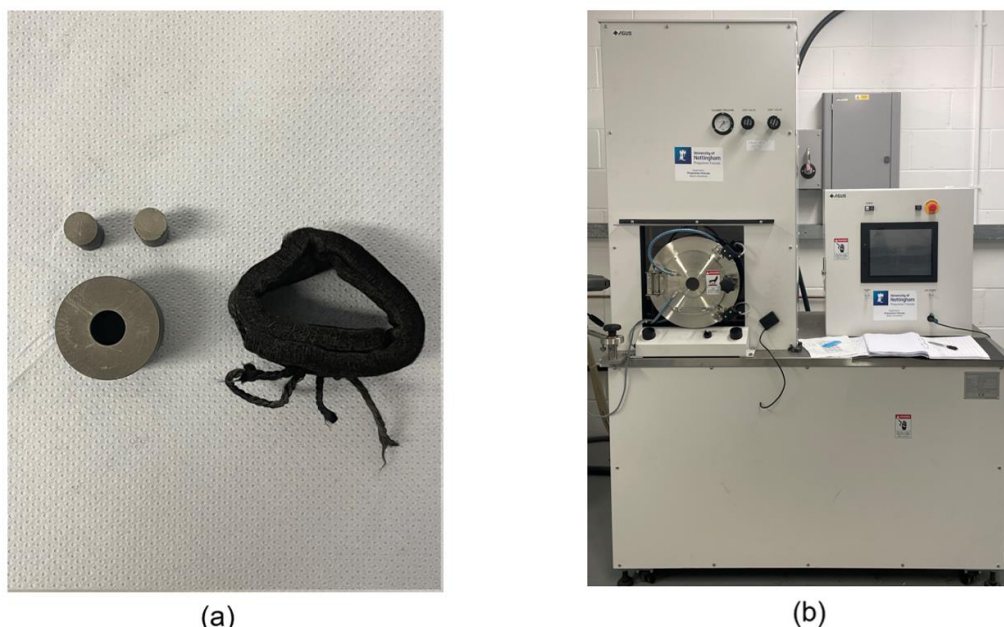


Figure 7. Images of: (a) graphite die, punches and cloth; (b) SPS equipment.

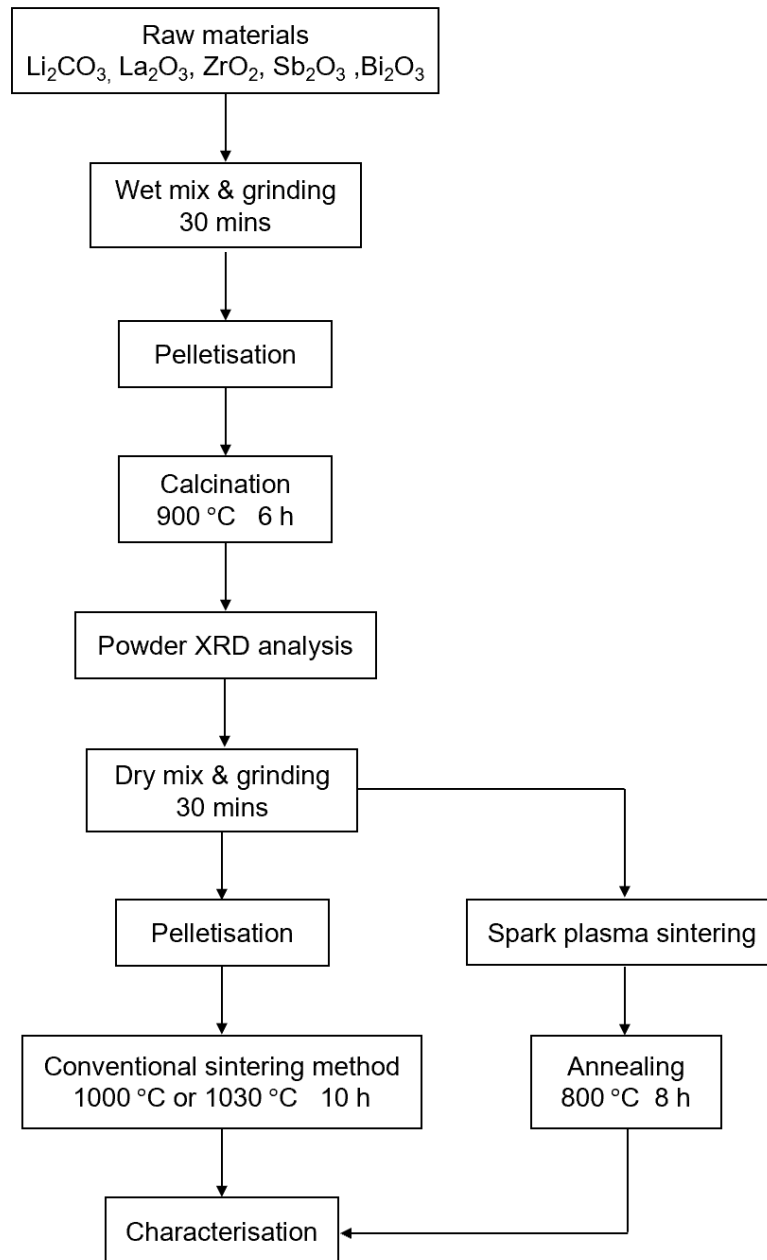


Figure 8. Flow chart of the LLZO synthesis

3.2. Characterisation

3.2.1. Density measurement

After synthesis, LLZO samples should be polished with sandpaper to make the surface smoother. Then, the densities of the pellets should be determined. The cross-sections of some samples were regular, but a few SPS sintered samples fractured after annealing due to internal residual stresses and became small irregular pieces. For regular-shaped samples, the mass-volume method was used to characterise the density. The mass of the samples was measured by the balance. The samples' thickness and diameter were measured by

calliper four times separately to calculate the mean value. After the measurement, the bulk density of LLZO samples can be calculated from the equation:

$$\rho = \frac{m}{V} = \frac{m}{t \cdot \frac{\pi}{4} D^2} \quad \text{Eqn.4}$$

where ρ (g/cm³) is the bulk density of LLZO sample, m (g) is the mass of the pellet, V (cm³) is the volume of the pellet, t (cm) is the thickness, D (cm) is the diameter.

For irregular-shaped samples, the surface area cannot be calculated from the measured diameter. Consequently, the density of the samples was measured using the Archimedes method. The sample was initially weighted in air and recorded as m_1 . It was then submerged in deionized water, and m_2 was recorded as its weight in water. The density of the water is 1 g/cm³. Then, the density of the samples can be calculated from the equation:

$$\rho = \frac{m_1 \cdot \rho_{\text{water}}}{m_1 - m_2} \quad \text{Eqn.5}$$

The theoretical density ρ_{th} of LLZO is obtained by refining the XRD data and the calculation from the equation:

$$\rho_{th} = \frac{N A_w}{V_c N_A} \quad \text{Eqn.6}$$

where N is the atoms number per unit cell, A_w (g/mol) is the molar mass of the atoms, V_c is the volume of unit cell and N_A is Avogadro's number (6.023×10^{23} atoms/mol). The theoretical density of LLZO is 5.098 g/cm³ [38].

Finally, the relative density ρ_{re} of the LLZO samples can be determined from the equation:

$$\rho_{re} = \frac{\rho}{\rho_{th}} \times 100\% \quad \text{Eqn.7}$$

3.2.2. X-ray diffraction (XRD)

The crystal structures and the phase purities of the LLZO samples were investigated by XRD using an X-ray diffractometer (Bruker D8 Advance, Germany) with Cu K α radiation ($\lambda = 1.5406$ Å). The image of the X-ray diffractometer is displayed in Figure 9. A step width of 0.02° was adopted for $2\theta = 10^\circ$ to 80° [16, 31]. The purpose of XRD was to analyse the influence of the doping element and the sintering conditions on the LLZO samples. The principle of XRD is that when a crystal is exposed to X-rays with a wavelength similar to the distance between atoms, it scatters an X-ray beam with the same wavelength. The resultant scattered waves lead to the creation of diffracted waves with higher intensities when they interfere. By analysing the spatial distribution of the diffracted intensity, it is possible to recognise the characteristics of the materials being examined. Bragg's approach is commonly used to explore diffraction by crystals. This technique involves treating crystals as being comprised of planes or layers. A portion of the X-rays is deflected off a plane at the same incidence angle, while the remainder continues to pass through to be subsequently

deflected by the succeeding planes [39]. For the XRD of a crystal, the correlation between the crystal structure and the diffraction lines should be demonstrated by Bragg's equation [40]:

$$n\lambda = 2d\sin\theta \quad \text{Eqn.8}$$

where n is arbitrary integer, λ is the wavelength of x-rays, d is the plane spacing between the adjacent atomic lattice, and θ is the angle between the x-rays and the scattering plane.



Figure 9. Image of the X-ray diffractometer.

3.2.3. Electrochemical impedance spectroscopy (EIS)

EIS (Figure 10) was used to test the impedance values of the LLZO samples, and then it can be used to determine the ionic conductivity of the samples. In this step, thin silver films were coated on the top and bottom surfaces of samples as Li-ion blocking electrodes. The ionic conductivity of samples can be calculated using EIS. The AC impedance measurement frequency range was 0.1 Hz to 1 MHz with an amplitude of 100 mV. The impedance was measured every 10 °C from room temperature to 90 °C. The samples were positioned at the centre of the quartz tube in the EIS equipment, and a K-type thermocouple was used to test the exact temperature [41]. Prior to the measurement, the samples should be thermally equilibrated at each temperature for approximately 10 minutes [42, 43].



Figure 10. Image of EIS equipment

The AC impedance plots (Nyquist plot) were fitted by Zview software to obtain the fitted data and equivalent circuit diagram. In Nyquist plot, the impedance is expressed in complex form. The horizontal axis $Z'(\Omega)$ represents the real component of the impedance Z , while the vertical axis $Z''(\Omega)$ represents the imaginary component of the impedance Z . The impedance plot was typically fitted into a semi-circle in the high-frequency region and a tail in the low-frequency region. The semicircle was contributed by the grain boundary and was represented as R_{gb} [41]. The tail Q_{el} represents the non-faradaic behaviour of the interface between the LLZO samples and the electrode. The total resistance R was determined by the x-coordinate of the intersection of the fitted tail and the horizontal axis. A typical Nyquist plot and the equivalent circuit are shown in Figure 11.

After obtaining the values of the resistance from the impedance plot, the ionic conductivity can be calculated according to the equation:

$$\sigma = \frac{t}{R \cdot S} \quad \text{Eqn.9}$$

where σ (S cm^{-1}) is the ionic conductivity, t (cm) is the thickness of the pellet, R (Ω) is the resistance and S (cm^2) is the surface area of the pellet.

In addition, the ionic conductivity is activated by temperature. The relationship between ionic conductivity, activation energy and temperature can be described by an Arrhenius-like temperature dependence [16, 44]:

$$\sigma(T) = A \exp\left(-\frac{E_a}{K_B T}\right) \quad \text{Eqn.10}$$

where A is the pre-exponential factor, E_a (eV) is the activation energy, K_B is the Boltzmann constant and T (K) is the absolute temperature.

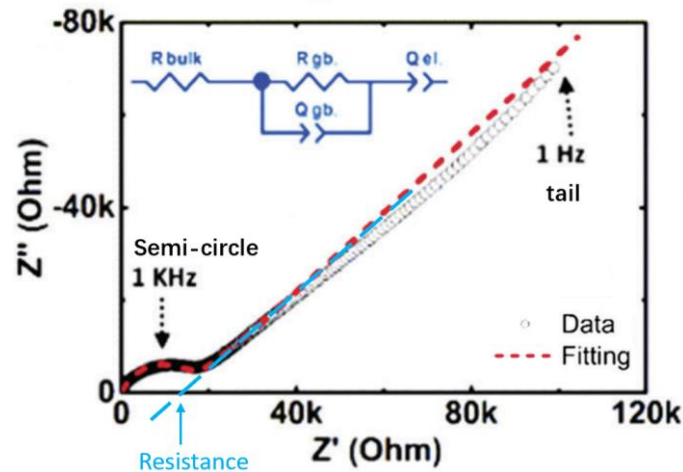


Figure 11. A typical Nyquist plot of LLZO and the equivalent circuit [41].

3.2.4. Scanning electron microscopy (SEM)

In the last stage, two pellets with relatively better performance ($\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ sintered at 1030 °C for 10 h and sintered at SPS 950 °C for 5 min) were selected to obtain the microstructure by SEM. Based on the same doping element with different sintering conditions, the effect of SPS techniques on the surface morphology can be analysed. The principle of SEM is that the electron microscope generates images by scanning the surface of the samples with a concentrated stream of electrons. These electrons interact with the atoms in LLZO samples and generate signals that include information about the surface topography and composition. A raster scan pattern is used by the electron beam, and the image is generated by the combination of the beam's position and the intensity of the signal [45].

4. RESULTS

0.0025 mol $\text{Li}_{6.6}\text{La}_3\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$, $\text{Li}_{6.6}\text{La}_{2.9}\text{Bi}_{0.1}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$, $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ and $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$, 0.005mol $\text{Li}_{6.6}\text{La}_{2.85}\text{Bi}_{0.15}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ were synthesised. The theoretical and actual mass of each raw material are shown in Table 2. Bi_2O_3 and Sb_2O_3 were used to achieve co-doping in LLZO. The Sb doping amount was fixed at 0.4, and only the doping amount of Bi and the sintering conditions were changed. Fifteen LLZO samples have been fabricated by the solid-state reaction method and SPS. Different sintering methods, temperatures and duration were used on the same composition to find the optimum sintering condition. The experimental data of these fifteen LLZO samples, including the area, thickness, ratio of area to thickness, and density are shown in Table 3. The experimental data consists of the ionic conductivity, sintering conditions and relative density are listed in Table 4.

All the samples can be divided into four sets of experiments to explore the performance of LLZO in detail. The aspects include the effect of Bi doping amounts on LLZO properties (Sample 1, 2, 6, 12), the effect of the sintering temperature of the conventional sintering method on LLZO properties (Sample 5, 6, 11, 12), the effect of different sintering techniques on LLZO properties (Sample 7, 8, 9, 10), and the effect of the Bi doping amount on the SPS techniques sintered LLZO properties (Sample 3, 4, 9, 14). The detailed result will be demonstrated in four sections in the following.

Table 2. Theoretical mass and actual mass of each raw powders

Composition	Mass of Li_2CO_3 (g)	Mass of La_2O_3 (g)	Mass of Sb_2O_3 (g)	Mass of Bi_2O_3 (g)	Mass of ZrO_2 (g)
Theoretical mass					
$\text{Li}_{6.6}\text{La}_3\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	1.3411	2.4436	0.2915	0	0.9858
$\text{Li}_{6.6}\text{La}_{2.9}\text{Bi}_{0.1}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	1.3411	2.3622	0.2915	0.1165	0.9858
$\text{Li}_{6.6}\text{La}_{2.85}\text{Bi}_{0.15}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	2.6822	4.6429	0.5830	0.3495	1.9716
$\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	1.3411	2.2807	0.2915	0.2330	0.9858
$\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	1.3411	2.1993	0.2915	0.3495	0.9858
Actual mass					
$\text{Li}_{6.6}\text{La}_3\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	1.3415	2.4433	0.2922	0	0.9858
$\text{Li}_{6.6}\text{La}_{2.9}\text{Bi}_{0.1}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	1.3418	2.3623	0.2917	0.1163	0.9857
$\text{Li}_{6.6}\text{La}_{2.85}\text{Bi}_{0.15}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	2.6825	4.6434	0.3498	1.9721	0.5831
$\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	1.3407	2.2807	0.2919	0.2331	0.9856
$\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	1.3411	2.1995	0.2914	0.3498	0.9859

Table 3. The values of area, thickness, ratio of area to thickness and density for all $\text{Li}_{6.6}\text{La}_{3-y}\text{Bi}_y\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ samples.

Sample No.	Composition	Sintering method, temperature and duration	Area (cm^2)	Thickness (cm)	A/t (cm)	Density (g/cm^3)
1	$\text{Li}_{6.6}\text{La}_3\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	1030°C 10h	0.8107	0.137	5.9175	3.40
2	$\text{Li}_{6.6}\text{La}_{2.9}\text{Bi}_{0.1}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	1030°C 10h	0.722	0.133	5.4286	4.29
3	$\text{Li}_{6.6}\text{La}_{2.9}\text{Bi}_{0.1}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	SPS 950°C 5min	0.3158	0.074	4.2676	4.78
4	$\text{Li}_{6.6}\text{La}_{2.85}\text{Bi}_{0.15}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	SPS 950°C 5min	0.2033	0.125	1.6264	4.84
5	$\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	1000 °C 10h	0.716	0.148	4.8378	4.0
6	$\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	1030°C 10h	0.7268	0.133	5.4647	4.05
7	$\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	SPS 750°C 5min	0.5924	0.136	4.3559	4.3
8	$\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	SPS 800°C 5min	0.4985	0.140	3.5607	4.63
9	$\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	SPS 950°C 5min	0.6185	0.129	4.7946	4.68
10	$\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	SPS 1000°C 3min	0.3297	0.144	2.2896	4.58
11	$\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	1000 °C 10h	0.7118	0.129	5.5178	4 .14
12	$\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	1030°C 10h	0.7103	0.126	5.6373	4.08
13	$\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	SPS 750°C 5min	0.431	0.122	3.5328	4.08
14	$\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	SPS 950°C 5min	0.4996	0.147	3.3986	4.54
15	$\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	SPS 1000°C 3min	0.2329	0.161	1.4466	4.65

Table 4. The ionic conductivity and relative density for all $\text{Li}_{6.6}\text{La}_{3-y}\text{Bi}_y\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ samples.

Sample No.	Composition	Bi	Sintering method, temperature and duration	Relative density (%)	Ionic conductivity at 30°C (S cm^{-1})
1	$\text{Li}_{6.6}\text{La}_3\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	0	1030°C 10h	66.7	1.79×10^{-8}
2	$\text{Li}_{6.6}\text{La}_{2.9}\text{Bi}_{0.1}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	0.1	1030°C 10h	84.1	6.65×10^{-5}
3	$\text{Li}_{6.6}\text{La}_{2.9}\text{Bi}_{0.1}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	0.1	SPS 950°C 5min	93.8	Not measurable
4	$\text{Li}_{6.6}\text{La}_{2.85}\text{Bi}_{0.15}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	0.15	SPS 950°C 5min	94.9	8.27×10^{-5}
5	$\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	0.2	1000 °C 10h	79.3	4.95×10^{-6}
6	$\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	0.2	1030°C 10h	79.5	2.28×10^{-5}
7	$\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	0.2	SPS 750°C 5min	84.4	1.13×10^{-5}
8	$\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	0.2	SPS 800°C 5min	90.9	1.09×10^{-5}
9	$\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	0.2	SPS 950°C 5min	91.7	3.12×10^{-5}
10	$\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	0.2	SPS 1000°C 3min	89.8	6.90×10^{-6}
11	$\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	0.3	1000 °C 10h	81.2	4.16×10^{-5}
12	$\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	0.3	1030°C 10h	80.2	1.75×10^{-4}
13	$\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	0.3	SPS 750°C 5min	80.0	Not measurable
14	$\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	0.3	SPS 950°C 5min	89.1	2.70×10^{-5}
15	$\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$	0.3	SPS 1000°C 3min	91.3	3.38×10^{-5}

4.1. Effects of Bi doping amount on the performance of LLZO samples

$\text{Li}_{6.6}\text{La}_{3-y}\text{Bi}_y\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ ($y=0, 0.1, 0.2, 0.3$) were fabricated under the identical sintering conditions (conventional sintering at 1030 °C for 10 h) to investigate the effect of Bi doping

concentration on the properties of LLZO. The samples were recorded as sample 1, 2, 6, 12 respectively (shown in Table 4). The result of this group of experiment will be divided into three parts to show the effect of Bi doping on crystal structure, ionic conductivity, activation energy and relative density.

4.1.1. Effects of Bi doping amount on the crystal structure of LLZO samples by XRD

Figure 12 illustrates the X-ray diffraction patterns of $\text{Li}_{6.6}\text{La}_{3-y}\text{Bi}_y\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ ($y=0, 0.1, 0.2, 0.3$) after 1030 °C conventional sintering for 10 h, to determine the effect of Bi doping amount on the crystal structure of LLZO. Figure 12 compares all the diffraction peaks to the standard cubic phase LLZO (PDF card 00-063-0174). It can be observed that all LLZO samples were well-matched to the standard cubic phase LLZO, confirming that the LLZO samples possessed cubic crystal structure. In addition, it can be found that the diffraction peak intensity of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ was significantly much higher than that of other samples. Also, an overall shift of the XRD patterns to the right is noticeable after doping the element Bi. What's more, some impurity phases are marked in Figure 12 including ZrO_2 , Li_2ZrO_3 and BiLi_3 . The reasons for the generation of these impurities will be analysed in Discussion section. In general, the crystal structure of the samples was improved after doping element Bi. Among these four samples, the crystal structure of $\text{Li}_{6.6}\text{La}_{2.9}\text{Bi}_{0.1}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ has the best phase purity, which is nearly 100% cubic phase.

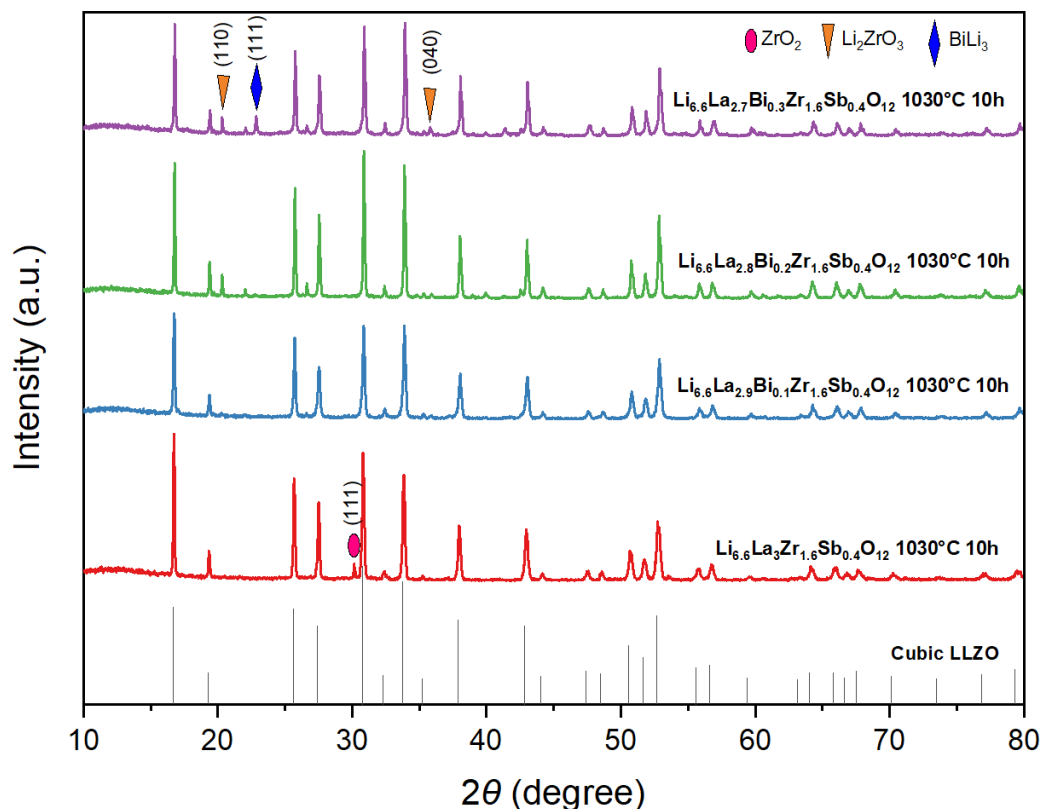


Figure 12. XRD patterns of $\text{Li}_{6.6}\text{La}_{3-y}\text{Bi}_y\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ ($y=0, 0.1, 0.2, 0.3$) sintered at 1030 °C for 10 h.

4.1.2. Effects of Bi doping amount on the ionic conductivity of LLZO samples by EIS

The ionic conductivity of $\text{Li}_{6.6}\text{La}_{3-y}\text{Bi}_y\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ ($y=0, 0.1, 0.2, 0.3$) samples was determined by the impedance. The impedance plots are demonstrated in Figure 13. A typical impedance plot displays one semicircle in the high-frequency range and a line in the low-frequency range. In Figure 13, only the impedance plot of $\text{Li}_{6.6}\text{La}_3\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ fits the above characteristics. The remaining three samples shown did not exhibit the above characteristics due to the substantial reduction of resistance after doping with Bi. Thus, an amplified AC impedance plot is needed to show the impedance of $\text{Li}_{6.6}\text{La}_{2.9}\text{Bi}_{0.1}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$, $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ and $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ (Figure 13b). The impedance plot of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ sample shows the arc and the inclined line clearly. It is noticeable that the impedance plots of $\text{Li}_{6.6}\text{La}_{2.9}\text{Bi}_{0.1}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ and $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ only depict inclined lines without clear semi-circles at the high frequency region. The response frequencies of the semicircles of $\text{Li}_{6.6}\text{La}_{2.9}\text{Bi}_{0.1}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ and $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ were out of 0.1 Hz to 1 MHz frequency range, so the semicircles cannot be detected. Compared to these four impedance plots, the resistance of $\text{Li}_{6.6}\text{La}_3\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ is approximately $7.2 \times 10^6 \Omega$ which is significantly larger than the other three samples. The resistance of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ is 8025.8Ω . The lowest resistance is possessed by $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ (1053.8Ω). The ionic conductivity measured at 30°C can then be calculated using Eqn.9, and the results are shown in Table 4.

Figure 14 illustrates the Arrhenius plot of $\text{Li}_{6.6}\text{La}_{3-y}\text{Bi}_y\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ ($y=0, 0.1, 0.2, 0.3$) as a function of temperature. The logarithm of the ionic conductivity can be fitted linearly with 1000 times the reciprocal of temperature. The activation energy can be estimated from the slope of the line between room temperature and 90°C which is according to Eqn.10. It is noticeable that the ionic conductivity was enhanced with the increase in temperature.

The variation of ionic conductivity and the activation energy of $\text{Li}_{6.6}\text{La}_{3-y}\text{Bi}_y\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ ($y=0, 0.1, 0.2, 0.3$) at 30°C are shown in Figure 15. As Figure 15 presents, the ionic conductivity of $\text{Li}_{6.6}\text{La}_3\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ is much lower ($1.79 \times 10^{-8} \text{ S cm}^{-1}$) at 30°C . It can be discovered that the ionic conductivity of LLZO is improved by almost three orders of magnitude after the incorporation of Bi. As Bi was increased to $y=0.1$, the ionic conductivity increased rapidly to $6.65 \times 10^{-5} \text{ S cm}^{-1}$, whereas the activation energy decreased significantly from 0.49 eV to 0.34 eV. When the content of Bi further increased to $y=0.2$, the ionic conductivity decreased to $2.28 \times 10^{-5} \text{ S cm}^{-1}$, but it was still much higher than the ionic conductivity of LLZO without Bi doping. Simultaneously, its activation energy increased from 0.36 eV to 0.45 eV. When the content of Bi was increased to 0.3, the sample possessed the highest ionic conductivity ($1.75 \times 10^{-4} \text{ S cm}^{-1}$) and the lowest activation energy (0.31 eV).

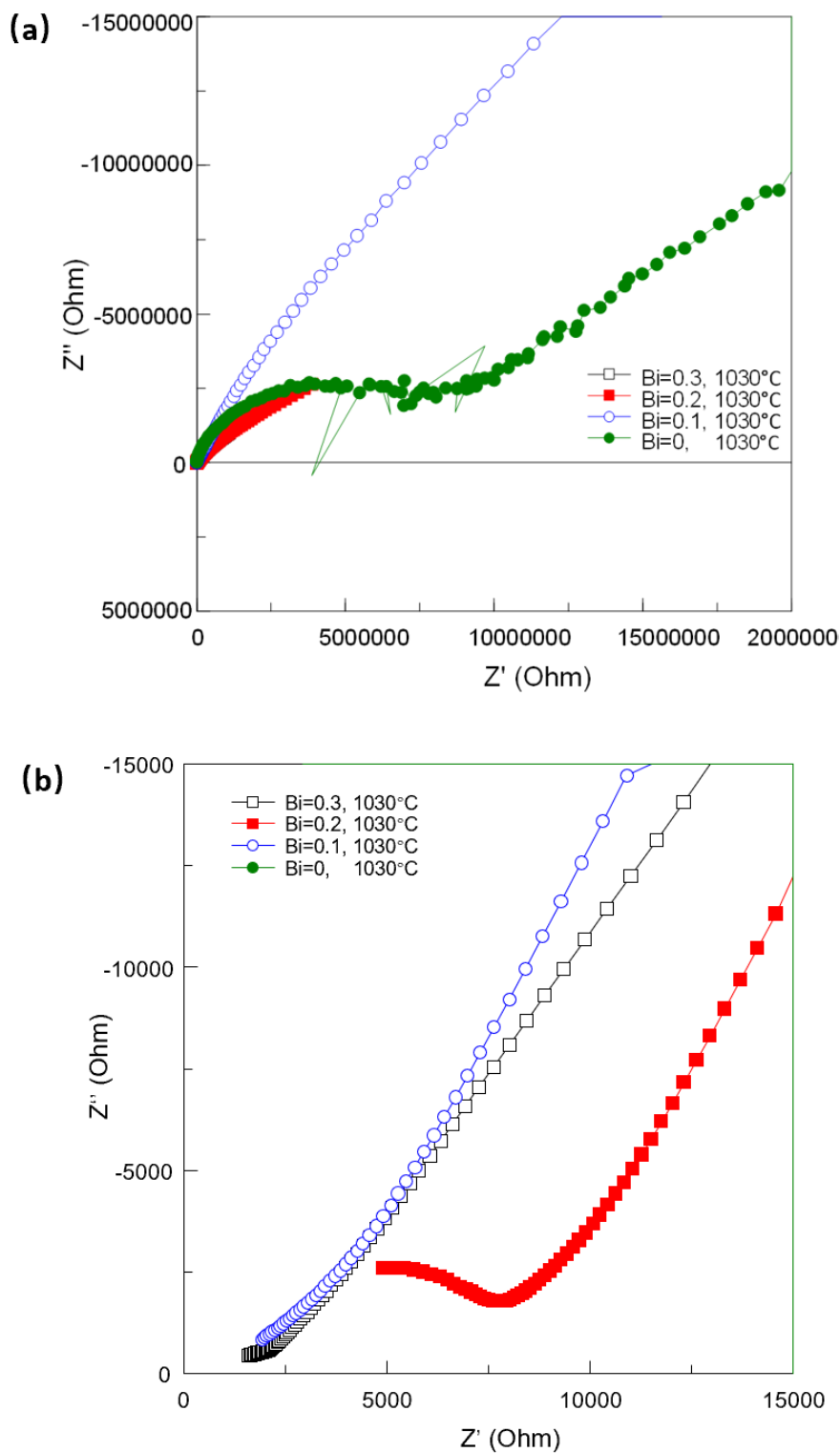


Figure 13. (a) Alternating current (AC) impedance plots of $\text{Li}_{6.6}\text{La}_{3-y}\text{Bi}_y\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ ($y=0, 0.1, 0.2, 0.3$) (sintered at 1030 °C for 10 h) measured at 30 °C; (b) Amplified AC impedance plots of $\text{Li}_{6.6}\text{La}_{3-y}\text{Bi}_y\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ ($y=0, 0.1, 0.2, 0.3$).

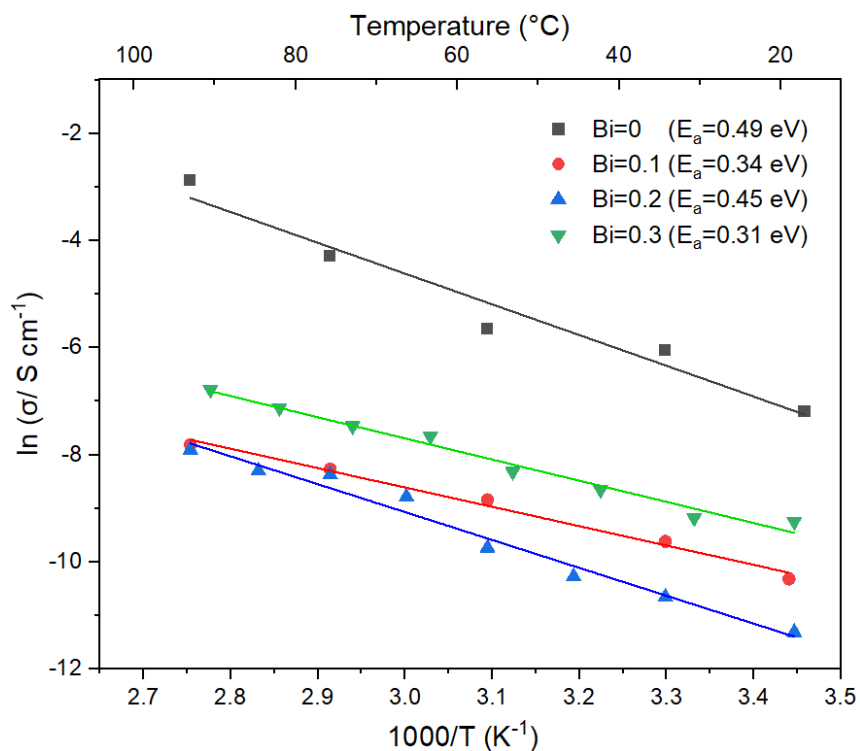


Figure 14. Arrhenius plot for the total ionic conductivity of $\text{Li}_{6.6}\text{La}_{3-y}\text{Bi}_y\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ ($y=0, 0.1, 0.2, 0.3$) sintered at 1030 °C for 10 h measured at 30 °C.

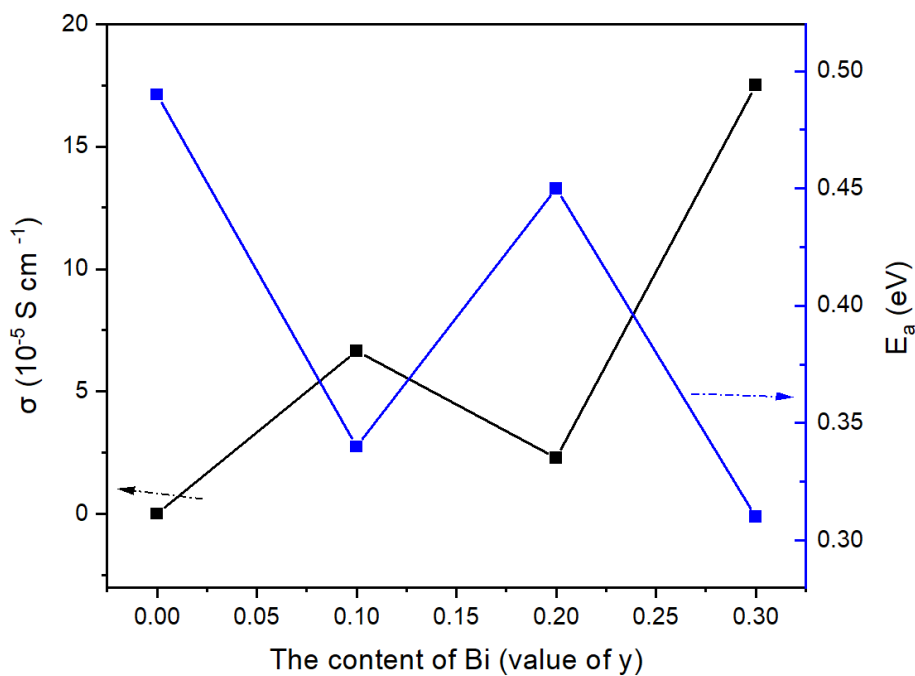


Figure 15. Total ionic conductivity measured at 30 °C, activation energy of $\text{Li}_{6.6}\text{La}_{3-y}\text{Bi}_y\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ ($y=0, 0.1, 0.2, 0.3$).

4.1.3. Effects of Bi doping amount on the relative density

The relative density of $\text{Li}_{6.6}\text{La}_{3-y}\text{Bi}_y\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ ($y=0, 0.1, 0.2, 0.3$) ceramics sintered at 1030 °C for 10 h is presented in Figure 16. $\text{Li}_{6.6}\text{La}_3\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ had the lowest relative density of 66.7%. After doping with the Bi element, the relative density significantly increased. When the Bi doping content was 0.1, the relative density reached the maximum value of 84.1%. This variation corresponded to the change in ionic conductivity. When the Bi concentration was further increased to 0.2 and 0.3, the relative density slightly decreased to 79.5% and 80.2%, respectively. These results demonstrate that a suitable amount of Bi doping can improve the relative density of the LLZO solid electrolytes effectively. In addition, the improved relative density also enhances ionic conductivity.

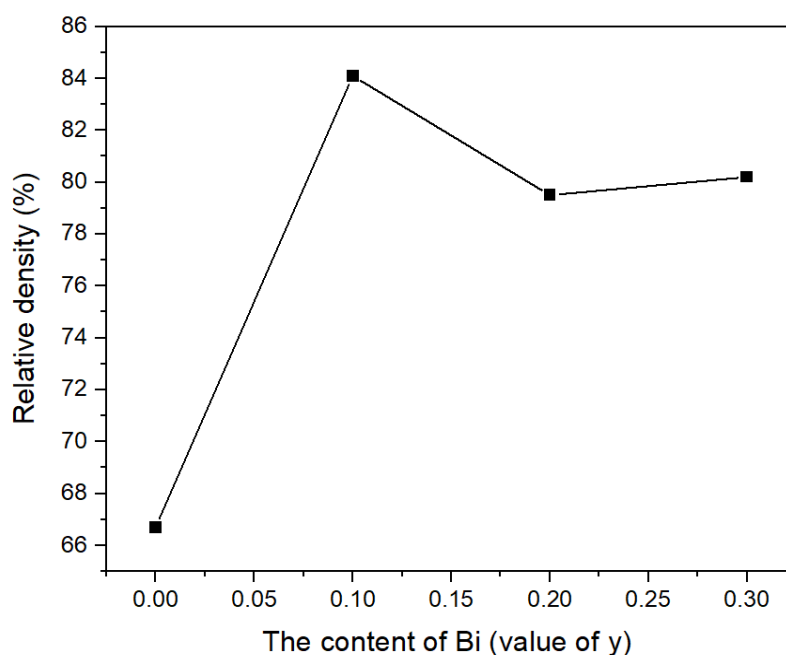


Figure 16. Relative density of $\text{Li}_{6.6}\text{La}_{3-y}\text{Bi}_y\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ ($y=0, 0.1, 0.2, 0.3$) ceramics sintered at 1030 °C for 10 h.

4.2. Effects of the sintering temperature on the performance of LLZO samples

$\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ and $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ were synthesised under different sintering temperature in this set of experiments to investigate the effect of conventional sintering temperature on the performance of LLZO samples. The sintering temperatures were set as 1000 °C and 1030 °C for 10 h. These four samples were recorded as sample 5, 6, 11, 12 respectively (shown in Table 4).

4.2.1. Effects of the sintering temperature on the crystal structure of LLZO

Figure 17 demonstrates the XRD patterns of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ and $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ ceramics sintered at 1000 °C and 1030 °C for 10 h. The XRD patterns of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ and $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ ceramics were all well

matched to the cubic LLZO. For $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$, the XRD patterns only indicate a minor change in the crystal structure at different temperatures. $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ electrolyte sintered at 1000 °C for 10 h has the impurity phase $\text{Li}_2\text{Zr}_2\text{O}_7$. This impurity peak diminished when the sintering temperature was increased to 1030 °C. $\text{Li}_2\text{Zr}_2\text{O}_7$ is the precursor of LLZO, the existence of $\text{Li}_2\text{Zr}_2\text{O}_7$ may be because the powder did not react sufficiently at 1000 °C. This could demonstrate that the sintering temperature of 1030 °C is optimal. This phenomenon did not display in $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ samples, indicating that $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ possibly had sufficient reaction at 1000 °C. In addition, the intensity of the Li_2ZrO_3 impurity diffraction peak is found to be stronger in $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ sintered at 1000 °C compared to the sample sintered at 1030 °C. This indicates that the Li_2ZrO_3 impurity concentration was higher in $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ sintered at 1000 °C.

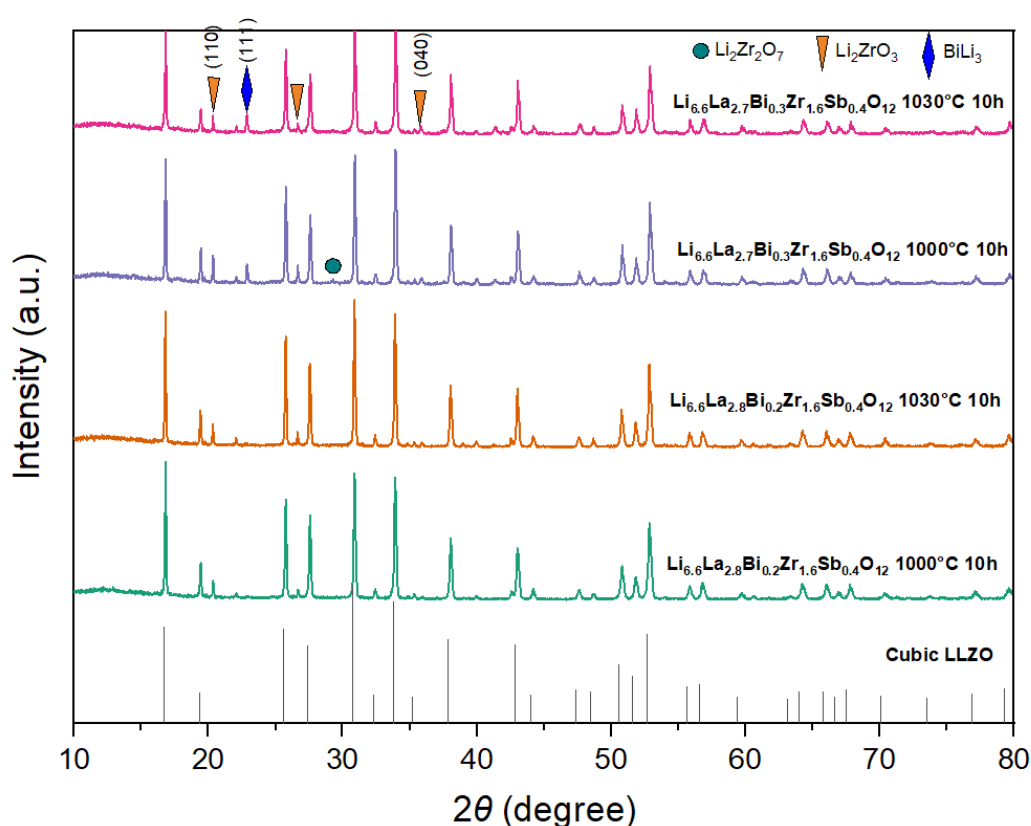


Figure 17. XRD patterns of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ and $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ ceramics sintered at 1000 °C and 1030 °C for 10 h.

4.2.2. Effects of the sintering temperature on the ionic conductivity of LLZO

EIS test was conducted to determine the impact of conventional method sintering temperature on LLZO's ionic conductivity. The impedance plots of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ and $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ ceramics sintered at two different temperatures are depicted in Figure 18. It can be discovered that the impedance spectra of two $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ samples are composed of clear arcs and inclined lines, while two $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$

samples lack distinct arcs at the high-frequency region. In addition, the resistance of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ sintered at 1030 °C is significantly lower than that of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ sintered at 1000 °C. The resistance of $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ sintered at 1030 °C is also lower than that of the sample sintered at 1000 °C. The ionic conductivity (measured at 30 °C) can then be calculated using Eqn.9 and the results are displayed in Table 4. For $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$, the ionic conductivity of the sample sintered at 1030 °C is $2.28 \times 10^{-5} \text{ S cm}^{-1}$ which is nearly an order of magnitude higher than that of the sample sintered at 1000 °C ($4.95 \times 10^{-6} \text{ S cm}^{-1}$). For $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$, the ionic conductivity of the solid electrolytes sintered at 1030 °C ($1.75 \times 10^{-4} \text{ S cm}^{-1}$) is also much higher than those of sintered at 1000 °C ($4.16 \times 10^{-5} \text{ S cm}^{-1}$). It is noticeable that when the temperature was increased from 1000 °C to 1030 °C, the ionic conductivity of both $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ and $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ were improved by nearly an order of magnitude.

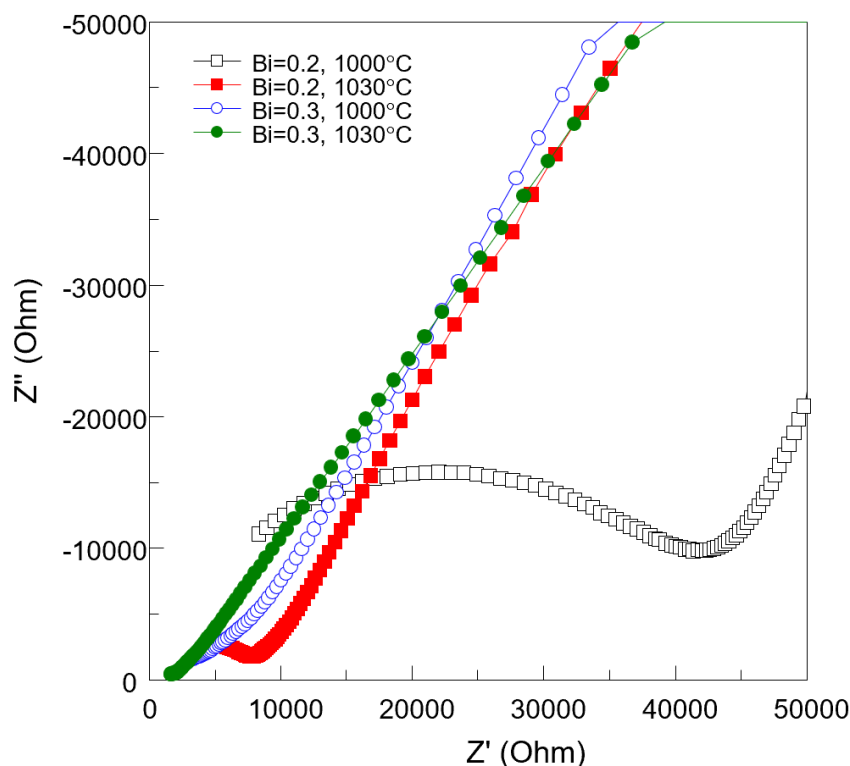


Figure 18. AC impedance plots of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ and $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ ceramics sintered at 1000 °C and 1030 °C for 10 h measured at 30 °C.

4.2.3. Effects of the sintering temperature on the relative density of LLZO

The values of the relative densities of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ and $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ ceramics sintered at 1000 °C and 1030 °C (Sample 5, 6, 11, 12) are shown in Table 4. It can be found that for the same compositions, the relative densities of the solid electrolytes sintered at 1000 °C and 1030 °C only have slight difference. For $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$, the relative density of the samples sintered at 1000 °C is nearly the same as that of samples

sintered at 1030 °C, they are 79.3% and 79.5% respectively. For $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$, the samples sintered at 1000 °C (81.2%) is slightly larger than the samples sintered at 1030 °C (80.2%).

4.3. Effects of different sintering technique on the performance of LLZO samples

It is reported that advanced sintering techniques can improve the performance of LLZO. One of the advanced sintering techniques, SPS, was used in the experiment to investigate its effect on the characteristics of LLZO. $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ powders were synthesised by SPS at 750 °C, 800 °C, 950 °C for 5 minutes, and 1000 °C for 3 minutes, respectively. These four samples were recorded as sample 7, 8, 9 and 10 respectively. For comparison, sample 6 fabricated by conventional sintering method was used to be analysed as well. In addition, it is noticeable that sample 10 used a different sintering duration from sample 7,8 and 9. This aimed to explore the effect of the SPS sintering time on the performance of LLZO samples.

4.3.1. Effects of different sintering techniques on the crystal structure of LLZO

The XRD patterns of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ samples sintered by conventional sintering methods and SPS at different temperatures are illustrated in Figure 19. This was done to find out how SPS affects the crystal structure of LLZO samples. Firstly, the diffraction peak intensity of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ sintered by SPS increased, and the diffraction peak was broadened compared to the sample sintered by conventional sintering. In addition, the XRD patterns of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ sintered by SPS at 750 °C, 800 °C, and 950 °C for 5 minutes are well matched to the cubic phase structure with some impurity phases. Li_2ZrO_3 was formed in both sintering techniques, but the impurity peak $\text{La}_2\text{Zr}_2\text{O}_7$ was only observed in the process of SPS. The detailed reason for forming these two impurities will be explained in the Discussion section. What's more, it is discovered that the cubic LLZO peaks intensity of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ sintered at 1000 °C for 3 minutes has a significant reduction in the cubic LLZO peak intensity due to the presence of numerous impurity phases. The impurity peak of Bi_2O_3 is observed, which is the initial reactant. Li_5BiO_5 and Li_5SbO_5 impurities are also labelled, and these are intermediate products in the reaction process. This phenomenon may be due to the insufficient reaction to fabricate LLZO under a 3-minute sintering duration. This indicates that the sintering time of 5 minutes for SPS is more appropriate, while 3 minutes is too short.

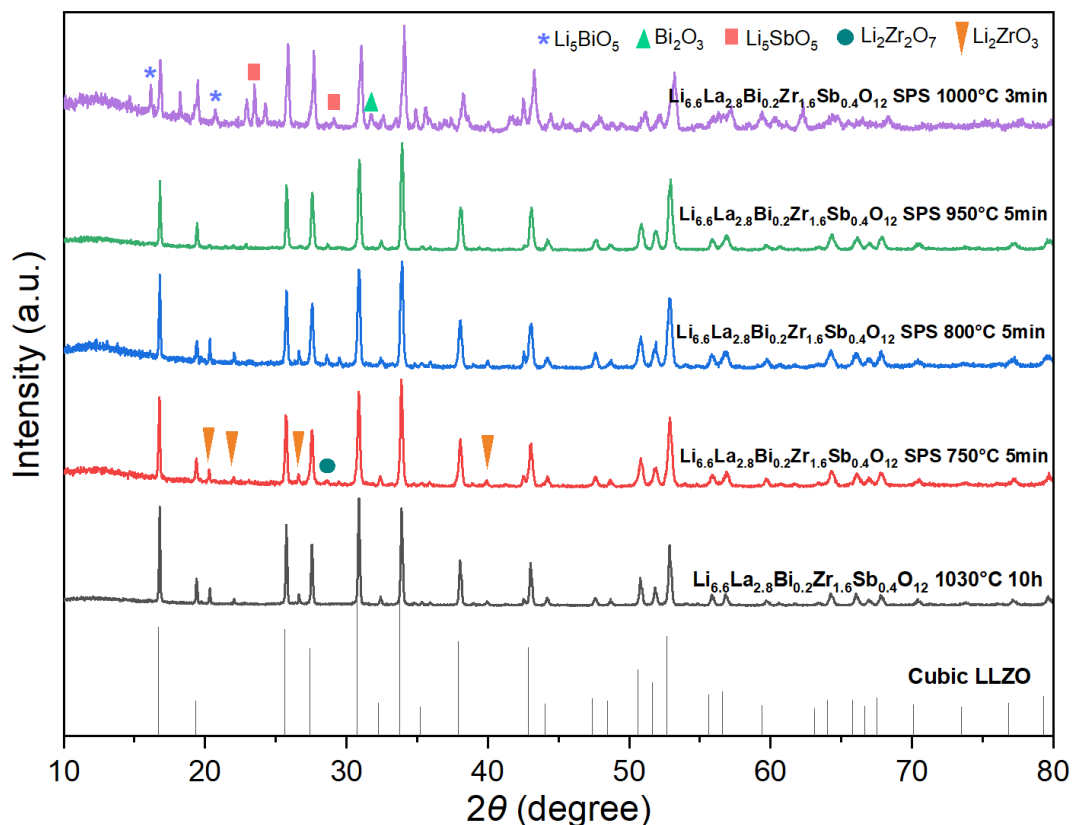


Figure 19. XRD patterns of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ samples sintered by SPS at different temperatures.

4.3.2. Effects of different sintering techniques on the ionic conductivity of LLZO

The impedance spectra of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ samples sintered by SPS at different temperatures were obtained by EIS (shown in Figure 20) to investigate the effect of SPS on the ionic conductivity of LLZO. It can be discovered that only $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ sintered at SPS 950 °C displays clear arcs and inclined lines, while the other three samples under different SPS temperatures do not have distinct arcs at the high frequency region. Among these four samples, the resistance of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ sintered at SPS 950 °C is the smallest (6590.4 Ω). According to Eqn.9, the calculated ionic conductivity is $3.12 \times 10^{-5} \text{ S cm}^{-1}$. This result is better than the ionic conductivity of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ sintered by the conventional sintering method. The calculated ionic conductivity of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ sintered at SPS 750 °C and 800 °C is $1.13 \times 10^{-5} \text{ S cm}^{-1}$ and $1.09 \times 10^{-5} \text{ S cm}^{-1}$ respectively, which is lower than the result of sample 9. This may be because the sintering temperature is too low. The ionic conductivity of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ sintered at SPS 1000 °C (Sample 10) is $6.90 \times 10^{-6} \text{ S cm}^{-1}$ which is much lower than the other three samples. This result is consistent with the cluttered crystal structure of Sample 10, probably due to the too short sintering duration.

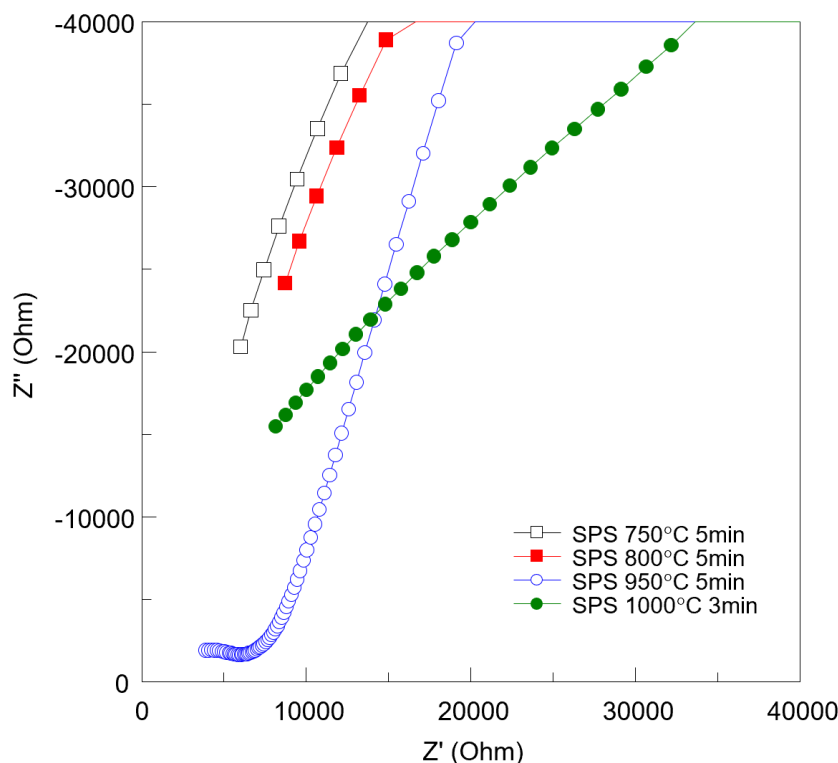


Figure 20. AC impedance plots of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ samples sintered by SPS at different temperature (measured at 30 °C).

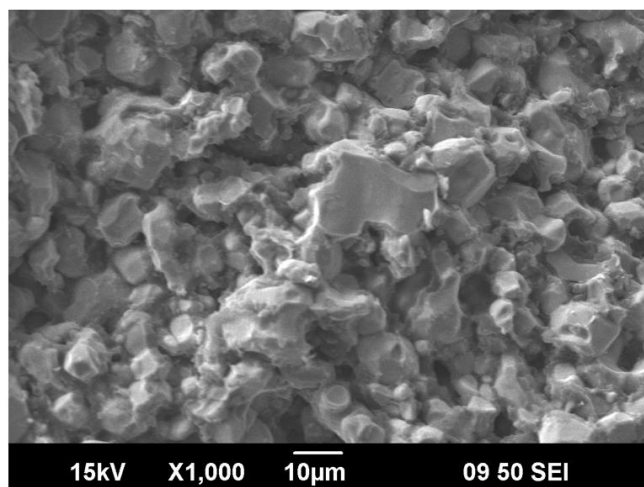
4.3.3. Effects of different sintering techniques on the relative density of LLZO

The results of the relative density of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ sintered by SPS at different temperatures are shown in Table 4. It can be discovered that the SPS technique can significantly improve LLZO samples' relative density compared to the samples sintered by the conventional method. $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ sample sintered at SPS 950 °C reached the highest relative density of 91.7%. Samples 8 and 10 had slightly lower relative densities, with 90.9% and 89.8%, respectively. $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ sample sintered at SPS 750 °C possessed the lowest relative density (84.4%) among these four SPS samples. This may be due to the sintering temperature being too low.

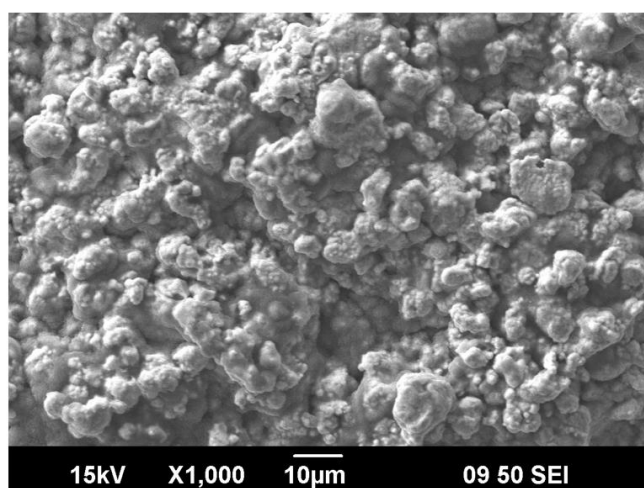
4.3.4. Effects of different sintering techniques on the morphologies of LLZO by SEM

The morphologies of the ceramic fracture sections of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ samples sintered by different techniques are shown in Figure 20. The grain size of the sample sintered by SPS (Figure 21b) is smaller than that of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ sintered by the conventional sintering method (Figure 21a). This result is consistent with the SPS sample's high relative density (91.7%), which is an extremely dense sample. This indicates that the SPS technique can inhibit the growth of grains. In addition, it is observable that the number

of pores on the grain boundaries of the samples sintered by the SPS technique was significantly reduced. Lower porosity may decrease the resistance of the grain boundaries in LLZO samples, which could lead to the higher ionic conductivity.



(a)



(b)

Figure 21. SEM images of the cross section of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$: (a) conventional sintering method at 1030 °C for 10 h; (b) SPS 950 °C for 5 min.

4.4. Effects of Bi doping amount under SPS method on the performance of LLZO samples

This section is set to investigate the effect of the Bi doping amount on the performance of SPS sintered LLZO samples. The sintering condition is fixed to SPS 950 °C for 5 min.

$\text{Li}_{6.6}\text{La}_{3-y}\text{Bi}_y\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ ($y=0.1, 0.15, 0.2, 0.3$) ceramics were synthesised and recorded as Sample 3, 4, 9, and 14 separately (shown in Table 4).

4.4.1. Effects of Bi doping amount on the crystal structure of SPS sintered LLZO

The XRD patterns of $\text{Li}_{6.6}\text{La}_{2.9}\text{Bi}_{0.1}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$, $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ and $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ sintered by SPS at 950 °C for 5 min are plotted in Figure 22. The crystal structures of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ and $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ sintered by SPS at 950 °C for 5 minutes are well matched to the cubic LLZO with some impurity phases. The impurity phase BiLi_3 also existed in $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$, which was due to the excessive Bi doping or the volatilized Li that reacted with Bi. This phenomenon appeared in both the SPS method and the conventional sintering method. Apart from this, it is discovered that the cubic LLZO peak intensity of $\text{Li}_{6.6}\text{La}_{2.9}\text{Bi}_{0.1}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ was reduced by many impurity phases. This result is quite different from the crystal structure of $\text{Li}_{6.6}\text{La}_{2.9}\text{Bi}_{0.1}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ synthesized by the conventional sintering method. $\text{Li}_{6.6}\text{La}_{2.9}\text{Bi}_{0.1}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ sintered by the conventional sintering method is almost 100% pure cubic LLZO phase.

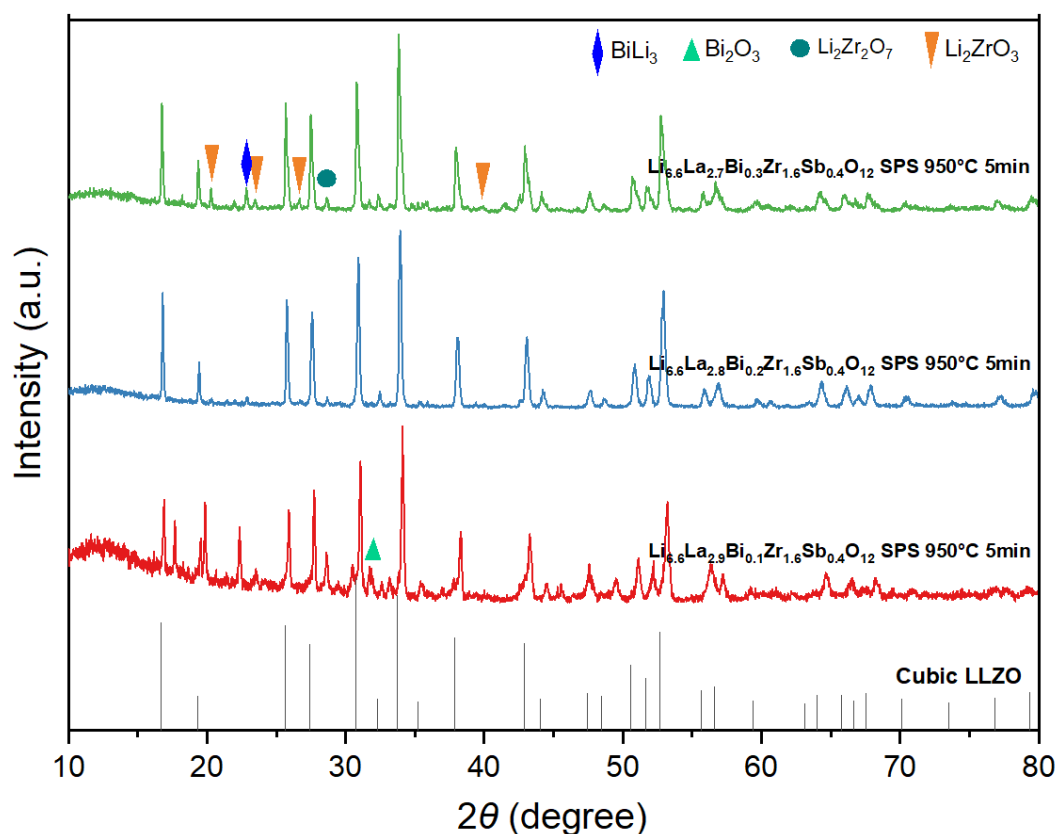


Figure 22. XRD patterns of $\text{Li}_{6.6}\text{La}_{3-y}\text{Bi}_y\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ ($y=0.1, 0.15, 0.2, 0.3$) (sintered at SPS 950 °C for 5 min).

4.4.2. Effects of Bi doping amount on the ionic conductivity of SPS sintered LLZO

The impedance spectra of $\text{Li}_{6.6}\text{La}_{3-y}\text{Bi}_y\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ ($y=0.1, 0.15, 0.2, 0.3$) sintered by SPS are shown in Figure 23. The arcs and inclined lines can be discovered in the impedance plots of $\text{Li}_{6.6}\text{La}_{2.85}\text{Bi}_{0.15}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ and $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$. The resistance of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ was the lowest (6590.4 Ω), while the resistance of

$\text{Li}_{6.6}\text{La}_{2.85}\text{Bi}_{0.15}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ was slightly higher (7431Ω). The resistance of $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ was much higher than that of these two samples. The resistance of $\text{Li}_{6.6}\text{La}_{2.9}\text{Bi}_{0.1}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ cannot be obtained because the impedance spectra do not conform to the typical impedance plot. The calculated ionic conductivity of $\text{Li}_{6.6}\text{La}_{2.85}\text{Bi}_{0.15}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ was the highest, which reached $8.27 \times 10^{-5} \text{ S cm}^{-1}$. With the increase in Bi content, the ionic conductivities of LLZO samples sintered by the SPS technique were gradually decreased to Bi=0.2 ($3.12 \times 10^{-5} \text{ S cm}^{-1}$) and Bi=0.3 ($2.7 \times 10^{-5} \text{ S cm}^{-1}$). This indicates that Bi=0.15 may be the optimal doping amount in SPS sintered LLZO, which is different from the optimal doping amount in LLZO sintered by the conventional sintering method.

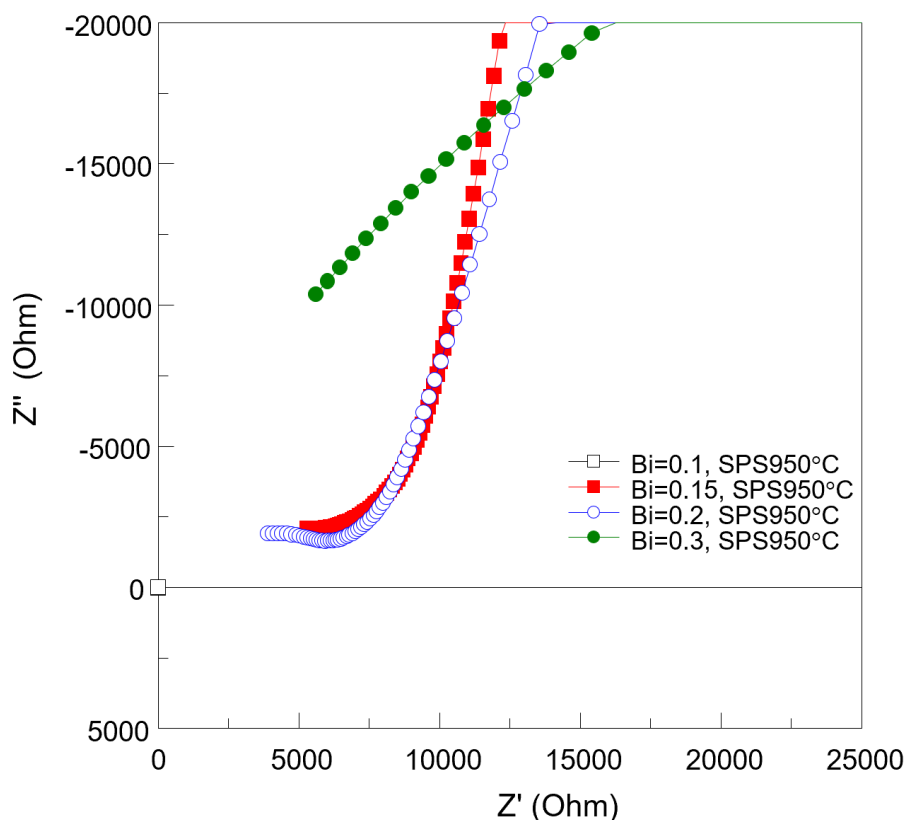


Figure 23. AC impedance plots of $\text{Li}_{6.6}\text{La}_{3-y}\text{Bi}_y\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ ($y=0.1, 0.15, 0.2, 0.3$) (sintered at SPS 950 °C for 5 min) measured at 30 °C.

4.4.3. Effects of Bi doping amount on the relative density of SPS sintered LLZO

The relative densities of Sample 3, 4, 9 and 14 are listed in Table 4. The comparison of the relative densities of LLZO with different Bi content sintered by different sintering methods is plotted (Figure 24). It is obvious that the relative densities of LLZO sintered by SPS are all significantly higher than those of the samples sintered by the conventional sintering method, indicating that the SPS technique can help improve the relative density of LLZO. For the LLZO sintered by SPS, the relative density reaches the highest at 94.9% when Bi=0.15. This

phenomenon corresponds to the highest ionic conductivity of this sample, which can demonstrate that the enhanced relative density can improve the ionic conductivity. When the Bi content is increased to 0.2 and 0.3, the relative density gradually decreases to 91.7% and 89.1%, separately.

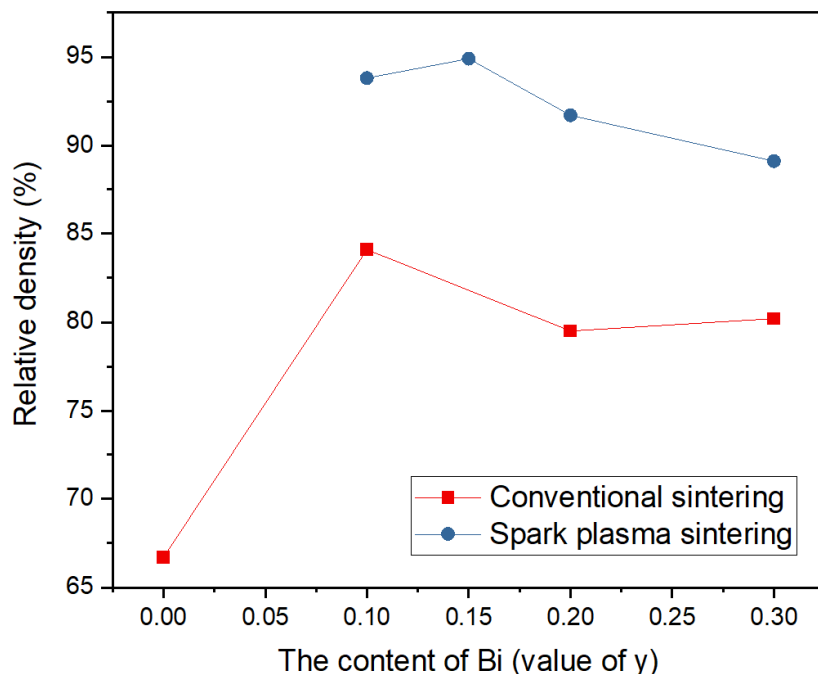


Figure 24. The comparison of the relative density of $\text{Li}_{6.6}\text{La}_{3-y}\text{Bi}_y\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ ($y=0, 0.1, 0.15, 0.2, 0.3$) ceramics sintered by different sintering method.

5. DISCUSSION

5.1. The analysis of effects of Bi doping amount on the performance of LLZO samples

In the first set of experiments, $\text{Li}_{6.6}\text{La}_{3-y}\text{Bi}_y\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ ($y=0, 0.1, 0.2, 0.3$) samples were prepared using the conventional sintering method at 1030 °C for 10 h to assess the effect of Bi doping amount on the performance of LLZO. The XRD analysis involved examining the intensity of the diffraction peak, peak broadening, peak asymmetries, peak shape, and whether the peak conforms to the standard card [46]. From the XRD patterns, the diffraction peaks of these samples exhibit the cubic LLZO phase. Different Bi-doping LLZO samples exhibit varying intensities of diffraction peaks. When Bi was doped into the LLZO samples, the crystallinity increased [47]. Additionally, the formation of impurity phases ZrO_2 , Li_2ZrO_3 and BiLi_3 were detected. ZrO_2 impurity phase was observed in $\text{Li}_{6.6}\text{La}_3\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$, which disappeared after doping with Bi due to incomplete raw material reactions during the solid-state reaction process. The impurity disappeared after Bi doping, demonstrating that Bi doping can contribute to a reduction in the sintering temperature or sintering time during the synthesis of c-LLZO to a certain extent [9]. In addition, the impurity phase Li_2ZrO_3 can be observed in $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ and $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$. This may be because the doping element Bi reduces the ideal sintering temperature, causing the reaction of the volatile Li with ZrO_2 . Apart from this, it is noticeable that the impurity peak BiLi_3 was generated in $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$. The reasons may be the excessive Bi doping, or the volatilized Li reacted with Bi. It indicated that the LLZO samples might become unstable at higher concentrations of Bi. Among these four samples, $\text{Li}_{6.6}\text{La}_{2.9}\text{Bi}_{0.1}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ is closest to the pure cubic phase. In general, Bi doping could improve the crystal structure of LLZO.

The resistance of LLZO samples can be derived from the AC impedance plot, along with the ionic conductivity and activation energy. Typically, an AC impedance plot consists of a semicircle in the high-frequency region and a straight line. However, it has been observed in practical experiments that the arc may not be visible. $\text{Li}_{6.6}\text{La}_3\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ and $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ samples show typical impedance plots, while $\text{Li}_{6.6}\text{La}_{2.9}\text{Bi}_{0.1}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ and $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ do not show clear arcs. Different interpretations exist for this phenomenon. Some research believes that the non-existence of a well-defined semicircle at high frequency suggests that the impedance caused by the grain boundary can be disregarded in comparison to that of the lattice [48]. Some research has given different opinions. It is argued that the absence of a discernible semicircle in samples with the highest electrical conductivity suggests that the resistance at the grain boundaries has almost disappeared [47].

Significant increases in ionic conductivity and relative density were observed following Bi doping. $\text{Li}_{6.6}\text{La}_{2.9}\text{Bi}_{0.1}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ has shown the optimum crystal structure, it also has the second highest ionic conductivity ($6.65 \times 10^{-5} \text{ S cm}^{-1}$), the highest relative density (84.1%), and the second lowest activation energy (0.34 eV). The reduction of activation energy is indicative of the decrease in the energy threshold for the migration of Li^+ and the increase in ionic conductivity [5]. Compared to $\text{Li}_{6.6}\text{La}_3\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$, it possessed the lowest ionic

conductivity ($1.79 \times 10^{-8} \text{ S cm}^{-1}$) and the lowest relative density (66.7%), as well as the highest activation energy of 0.49 eV. This change could be interpreted by the significantly increased relative density, phase purity, and occupation of 24c sites by La^{3+} ions [9]. In addition, this change can demonstrate that Bi^{3+} can improve the transportation properties of Li^+ in the lattice [49]. The performance displayed by the sample $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ was inferior to that of $\text{Li}_{6.6}\text{La}_{2.9}\text{Bi}_{0.1}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$, manifested as a decrease in ionic conductivity to $2.28 \times 10^{-5} \text{ S cm}^{-1}$, an increase in activation energy to 0.45 eV, and a drop in relative density to 79.5%. Additionally, the impurities in the crystal structure of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ increased. For $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$, the relative density (80.2%) was inferior to that of $\text{Li}_{6.6}\text{La}_{2.9}\text{Bi}_{0.1}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$, and the XRD pattern indicated the possibility of excessive Bi doping. However, the ionic conductivity reached the highest ($1.75 \times 10^{-4} \text{ S cm}^{-1}$) and the activation energy was the lowest (0.31 eV). This result could be attributed to errors in the measurement of the relative density. In addition to this, it is reported that the pellet's thickness may influence the ionic conductivity [32]. Consequently, this result may be caused by the distinct thickness of these four pellets.

Thus, it can be concluded that the ionic conductivity and relative density could be significantly enhanced by Bi element doping, with the optimal Bi doping concentration possibly being 0.3. Simultaneously, Bi doping can effectively reduce the sintering temperature of forming c-LLZO, thereby considerably decreasing the energy needed to produce the solid electrolytes and mitigating the challenges of the LLZO application [9].

5.2. The analysis of effects of sintering temperature on the performance of LLZO samples

According to reports, the sintering temperature significantly affects the densification of LLZO. For LLZO solid electrolytes, the appropriate sintering temperature could determine whether it is a tetragonal or cubic phase, and the phase can largely determine the ionic conductivity value [5]. In the second set of experiments, $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ and $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ were sintered at different temperatures (1000 °C and 1030 °C) for 10 h to investigate the influence on the performance of LLZO, and to find the optimum sintering temperature. According to XRD patterns, the samples sintered at these two different temperatures exhibited the cubic phase. The comparison of the two $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ samples sintered at different temperatures revealed an enhancement in both the ionic conductivity and the relative density. From the impedance plots of two $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ samples, it is obvious that the resistance decreased considerably as the temperature increased from 1000 °C to 1030 °C. Consequently, the ionic conductivity increased significantly from $4.95 \times 10^{-6} \text{ S cm}^{-1}$ to $2.28 \times 10^{-5} \text{ S cm}^{-1}$. The relative density only had a slight improvement, increasing by only 0.2% (from 79.3% to 79.5%). Compared to two $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ samples sintered at different temperatures, there was an improvement in the ionic conductivity (from $4.16 \times 10^{-5} \text{ S cm}^{-1}$ to $1.75 \times 10^{-4} \text{ S cm}^{-1}$) with the temperature increase from 1000 °C to 1030 °C. However, the

change in relative density did not coincide with the ionic conductivity change. When the temperature was higher, the relative density decreased from 81.2% to 80.2%. Errors in the calculation of the relative density may be responsible for this atypical change. During the fabrication process of the samples, a flaw may be generated, leading to errors in the measurement of the mass and the volume of the pellets, resulting in an inaccurate density calculation. From the XRD patterns of $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ at different temperatures, it can be observed that the impurity peak $\text{Li}_2\text{Zr}_2\text{O}_7$ disappeared with the increasing temperature. The emergence of $\text{Li}_2\text{Zr}_2\text{O}_7$ may be attributed to the incomplete reaction of the powders at 1000 °C, which suggests that the optimal sintering temperature may be around 1030 °C.

The results of the variation of the ionic conductivity with temperature are consistent with the findings obtained by Wood et al., the ionic conductivity improves progressively with increased sintering temperature [50]. The lower ionic conductivity at 1000 °C may be a result of the partial sintering of the pellets and an increase in porosity, which increases the grain boundary resistance [50]. The increased sintering temperature can facilitate the growth of the LLZO crystals, improve the quality of the LLZO grain boundaries, and enhance the integrity of the crystal structure to enhance the ionic conductivity [51]. Additionally, higher sintering temperatures can eliminate some impurities phase to increase the ionic conductivity. However, if the sintering temperature is too high, it may cause an increment in the grain size of LLZO, resulting in increased resistance and decreased ionic conductivity [51]. This set of experiments only compared two temperatures and cannot strongly support the claim that increasing the sintering temperature can increase LLZO conductivity. Conversely, lowering the temperature to some extent can prevent lithium loss because it can mitigate the influence of the high temperature on the interfacial reaction between the solid electrolytes and cathodes [52].

To summarise the findings in this set of experiments, 1030 °C sintering temperature is optimum to obtain the stable c-LLZO. The ionic conductivity and the relative density can be improved by increasing the sintering temperature from 1000 °C to 1030 °C.

5.3. The analysis of effects of different sintering technique on the performance of LLZO samples

This set of experiments was conducted to investigate the effect of the advanced sintering technique SPS on the characteristics and performance of LLZO. $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ were sintered by SPS at different temperatures (750 °C, 800 °C, 950 °C, and 1000 °C) to find the optimum SPS temperature. Simultaneously, different sintering durations were tried to investigate the effect of SPS holding time on the performance of LLZO. Several important points can be made from XRD patterns. Firstly, compared to $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ sintered by conventional sintering method, the diffraction peaks intensity of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ sintered by SPS at 750 °C, 800 °C, and 950 °C for 5 minutes increased. This indicates that SPS may increase the crystallinity of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$

samples. In addition, it is discovered that the diffraction peaks of the samples synthesised by SPS were broadened. In the process of SPS, microcracks were generated on the surface of the pellets to release internal stress. At this time, a large strain existed on the surface of the microcracks, which may lead to the diffraction peak being wider. Although the pellet was annealed after SPS, it may be inadequate which did not avoid the internal residual stresses in the pellets. The reason may also be the imperfect crystal lattice and chemical heterogeneities [46]. The shifting and broadening of the diffraction peak can be attributed to the generation of a distorted garnet phase [53]. Thirdly, Li_2ZrO_3 formed during the SPS process. This may be caused by the excess Li and the burial sintering in mother powder. Li_2ZrO_3 may not influence the ionic conductivity significantly because it is a conductive phase of lithium [5]. It is noticeable that the impurity peak $\text{La}_2\text{Zr}_2\text{O}_7$ was generated in the process of SPS but did not exist in the sample sintered by the conventional sintering method. A possible cause of $\text{La}_2\text{Zr}_2\text{O}_7$ formation is Li volatilization during the SPS process. $\text{La}_2\text{Zr}_2\text{O}_7$ is a non-conductive phase that may impact ionic conductivity [5].

According to the relative density results, the relative density of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ sintered by SPS was significantly larger than that of the conventional sintered sample. The sample sintered at SPS 950 °C obtained the highest relative density of 91.7%. This result corresponds to the enhanced ionic conductivity ($3.12 \times 10^{-5} \text{ S cm}^{-1}$), which is also the highest in this set compared to the ionic conductivity of the conventional sintering sample ($2.28 \times 10^{-5} \text{ S cm}^{-1}$). Simultaneously, the reason for the enhanced ionic conductivity can be obtained from the SEM microstructure. From SEM morphologies, it is noticeable that the SPS technique can lead to a reduction in the internal pore space and the shrinkage of the grain size of the sample [5]. It was discovered that the relative density was improved gradually with the increase of the SPS temperature from 750 °C to 950 °C (84.4% at 750 °C, 90.9% at 800 °C). This variation was consistent with the change in ionic conductivity. Based on these two characteristics, it can be indicated that SPS 950 °C may be the optimum SPS temperature. However, as the temperature rose from 950 °C to 1000 °C, the relative density slightly decreased to 89.8%, and the ionic conductivity significantly decreased to $6.9 \times 10^{-6} \text{ S cm}^{-1}$. The sintering duration of the SPS at 1000 °C was only 3 minutes, and in combination with the mess XRD pattern analysis, the cause was insufficient reaction due to the too short sintering time. It was confirmed that 5 minutes might be an acceptable sintering duration for SPS. Additionally, it is noticeable that SPS cannot always effectively improve the performance of LLZO if the sintering temperature and time are unsuitable. This optimal condition necessitates extensive testing to discover.

Therefore, it can be concluded that SPS can effectively optimise the performance of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ solid electrolytes under the appropriate sintering temperature and sintering duration. The suitable SPS temperature is approximately 950 °C, and the sintering time is close to 5 minutes.

5.4. The analysis of effects of Bi doping amount under SPS method on the performance of LLZO samples

In the initial set of experiments, the optimum doping ratio of Bi obtained via conventional sintering was around 0.3. In the third set of experiments, the advanced sintering technique SPS could effectively improve the performance of LLZO with the best sintering condition at SPS 950 °C for 5 min. Thus, this set was generated to investigate the effects of the Bi doping ratio on the SPS sintered LLZO. $\text{Li}_{6.6}\text{La}_{3-y}\text{Bi}_y\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ ($y=0.1, 0.15, 0.2, 0.3$) samples were synthesised by SPS at 950 °C. Compared to the samples sintered by SPS and those sintered by the conventional sintering method with the same composition, the relative densities were all significantly improved. This may be due to the different mechanisms of the SPS process, which applied high pressure and an electric field. Consequently, LLZO solid electrolytes could form dense grain boundaries during the SPS process. It may also be attributed to the densification and the combination of grains promoted by the surface activation, which is generated by the electric field [54-56].

By analysing the conductivity and relative density of the SPS sintered samples as a function of Bi content, it was found that as the Bi doping content increased from 0.1 to 0.15, the relative density increased and reached a maximum value (from 93.8% to 94.9%) in all samples. Simultaneously, the ionic conductivity also reached its highest at $8.27 \times 10^{-5} \text{ S cm}^{-1}$. It should be noted that the ionic conductivity of sample $\text{Li}_{6.6}\text{La}_{2.9}\text{Bi}_{0.1}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ sintered at SPS 950 °C was not measurable. It may be due to the thickness of the pellet. It was found that the thickness of this pellet was much smaller than other samples [32]. In addition, according to the XRD pattern, it may also be due to too many impurities. When the Bi doping amount was increased further, the ionic conductivity and the relative density gradually decreased. The ionic conductivity of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ decreased to $3.12 \times 10^{-5} \text{ S cm}^{-1}$, and that of $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ further decreased to $2.70 \times 10^{-5} \text{ S cm}^{-1}$. Similarly, the relative density of $\text{Li}_{6.6}\text{La}_{2.8}\text{Bi}_{0.2}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ decreased to 91.7% and then further decreased to 89.1%. From these results, it can be determined that the optimum Bi doping amount of the SPS sample is around 0.15, which is different from the optimum doping amount of the sample sintered by the conventional method. Under the optimum doping amount, the ionic conductivity of the SPS sample (8.27×10^{-5}) is lower than that of the conventional sintered sample ($1.75 \times 10^{-5} \text{ S cm}^{-1}$).

Thus, it can be concluded that the variation of LLZO performance with Bi doping amount under SPS differs from that under conventional sintering. The optimum doped Bi content under SPS is around 0.15.

6. CONCLUSIONS

In this lab-based individual project, fifteen $\text{Li}_{6.6}\text{La}_{3-y}\text{Bi}_y\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ ($y=0, 0.1, 0.15, 0.2, 0.3$) samples were synthesised by different sintering conditions. All the samples have successfully achieved the cubic LLZO structure. Several important findings could be summarised. Firstly, Bi-doped LLZO samples with different doping amounts show an increase in ionic conductivity of three orders of magnitude compared to the undoped sample under the same sintering method. The highest ionic conductivity of LLZO samples was achieved by $\text{Li}_{6.6}\text{La}_{2.7}\text{Bi}_{0.3}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ sintered at 1030 °C for 10 h ($1.75 \times 10^{-4} \text{ S cm}^{-1}$), which is also the only sample to reach $10^{-4} \text{ S cm}^{-1}$. Under SPS method, the optimal Bi doping amount is different. When the content of Bi is 0.15, the ionic conductivity and the relative density reached the highest value of $8.27 \times 10^{-5} \text{ S cm}^{-1}$ and 94.7 %, respectively. This demonstrates that the optimum content of doped Bi is probably 0.15 and, and 950 °C is the optimum SPS temperature. The relative density of $\text{Li}_{6.6}\text{La}_{2.85}\text{Bi}_{0.15}\text{Zr}_{1.6}\text{Sb}_{0.4}\text{O}_{12}$ is also the highest among the fifteen samples. Secondly, sintering temperature also affects the ionic conductivity of LLZO. Under conventional sintering method, the performance of samples sintered at 1030 °C is better than that of samples sintered at 1000 °C, indicating that the optimum sintering temperature is around 1030 °C. Furthermore, it is confirmed that SPS could improve the performance of LLZO under the appropriate sintering temperature and sintering time. The relative density of LLZO sintered by SPS could be significantly increased.

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APPENDIX 1– Health and Safety documents

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